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System and method for machine learning and augmented reality based user application

Abstract

The invention synthesizes a social network, electronic commerce, an intelligent (self-learning) subsystem (that may include a digital personal assistant (DPA) and/or an autonomous software agent, which can learn, adapt and take (meaningful) autonomous actions to solve one or more problems on a user's behalf) and a machine learning algorithm(s) or an artificial neural network (ANN). The synthesized social commerce integrates/utilizes stored information and near real time information/data/image(s) from an object/array of objects (Internet of Things (IoT)) and autonomous software agents. The machine learning algorithm(s) or the artificial neural network (ANN) can include (i) natural language processing (NLP) and (ii) a transformer model or a diffusion model.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is [0002] a continuation-in-part (CIP) patent application of (a) U.S. Non-Provisional patent application Ser. No. 18/831,206 entitled "INTELLIGENT (SELF-LEARNING) SUBSYSTEM IN ACCESS NETWORKS, filed on Sep. 14, 2024, wherein (a) a continuation-in-part (CIP) of patent application of (b) U.S. Non-Provisional patent application Ser. No. 18/445,647 entitled "INTELLIGENT (SELF-LEARNING) SUBSYSTEM IN ACCESS NETWORKS", filed on Dec. 9, 2023 (which resulted in a U.S. Pat. No. 12,101,609, issued on Sep. 24, 2024), [0003] a continuation-in-part (CIP) patent application of (c) U.S. Non-Provisional patent application Ser. No. 17/803,756 entitled "SYSTEM AND METHOD FOR MACHINE LEARNING AND AUGMENTED REALITY BASED USER APPLICATION", filed on Nov. 14, 2022. [0004] The entire contents of all (i) U.S. Non-Provisional patent applications, (ii) U.S. Provisional patent applications, as listed in the previous paragraph and (iii) the filed (patent) Application Data Sheet (ADS) are hereby incorporated by reference, as if they are reproduced herein in their entirety.

FIELD OF THE INVENTION
[0005] The present invention generally relates to the integration of a social network, electronic commerce (including performance based advertisement and electronic payment) and a mobile internet device (MID)/intelligent (self-learning) subsystem. The intelligent (self-learning) subsystem can be a mobile device or a premise device or a wearable device.
[0006] Furthermore, synthesized social electronic commerce dynamically integrates stored information, information (preferably real time information), communication with an object/array of objects (where the object can be coupled with a wireless (or radio) transmitter and/or a sensor)—Internet of Things (IoT) and a unified algorithm.
[0007] The unified algorithm can include a software agent, a fuzzy logic algorithm, a predictive algorithm, an intelligence rendering algorithm and a self-learning (including relearning) algorithm. It should be noted that real time information is near real time information in practice.
BACKGROUND OF THE INVENTION
[0008] Social networking is no longer just about making social connections online. User experience can be enhanced not only by connecting with people, but also by connecting with information (preferably real time information) and communicating with the object/array of objects.
[0009] The cornerstone of today's electronic commerce is based on converting a probable click (in a search engine) into an actual sale.
[0010] By synthesizing social networking with the electronic commerce, one can deliver consistent user experience across all touch-points (e.g., social, mobile and in-store).
[0011] Furthermore, synthesized social electronic commerce can also integrate stored information, real time information, data from the mobile internet device/intelligent (self-learning) subsystem and real time information/data/image(s) from the object/array of objects, where the object can be coupled with a wireless (or radio) transmitter and/or a sensor.
[0012] However, the mobile internet device/intelligent (self-learning) subsystem can preferably

communicate with a node, which can further communicate with the object/array of objects for spatial and time averaged information/data/image(s) from the object/array of objects.
[0013] The integration of social networking with (real time) user location information from a user's mobile internet device/intelligent (self-learning) subsystem and information/data/image(s) from the object/array of objects can embed physical reality into an internet space and an internet reality into a physical space.
[0014] Furthermore, the unified algorithm (integrating a software agent, a fuzzy logic algorithm, a predictive algorithm, an intelligence rendering algorithm and a self-learning (including relearning) algorithm can add a new dimension to user experience.
[0015] Furthermore, by designing a system-on-chip (SoC) (an advanced microprocessor integrated with a security algorithm) or a Super System on Chip (SSoC) for the mobile internet device/intelligent (self-learning) subsystem, intelligence can also be rendered to the mobile internet device/intelligent (self-learning) subsystem.
SUMMARY OF THE INVENTION
[0016] The invention synthesizes the social network, electronic commerce (including the performance based advertisement and electronic payment) and mobile internet device/intelligent (self-learning) subsystem (intelligence is achieved by utilizing advanced algorithm(s) and/or advanced microprocessor design(s) for the mobile internet device/intelligent (self-learning) subsystem).
[0017] The synthesized social electronic commerce further dynamically integrates stored information, real time information, information/data/image(s) from the object/array of objects (where the object can be coupled with a wireless (or radio) transmitter and/or a sensor) and the unified algorithm (which includes a software agent, a fuzzy logic algorithm, a predictive algorithm, an intelligence rendering algorithm and a self-learning (including relearning) algorithm).

Description

BRIEF DESCRIPTION OF THE FIGURES
[0018] FIGS. **1**A and **1**B illustrate an integrated application flow chart of a social wallet, according to one embodiment of the present invention.
[0019] FIGS. **2**A, **2**B, **2**C, **2**D, **2**E, **2**F, **2**G and **2**H illustrate a method flow chart of integration of the social wallet, electronic commerce and performance based advertisement, according to one embodiment of the present invention.
[0020] FIG. **3**A illustrates a block diagram of a social wallet electronic module, according to one embodiment of the present invention.
[0021] FIGS. **3**B and **3**C illustrate a block diagram of an application of a biosensor (as a sensor) of the social wallet electronic module, according to one embodiment of the present invention.
[0022] FIGS. **3**D, **3**E and **3**F illustrate cross-sections of various configurations of a solar cell (an electrical power provider component) of the social wallet electronic module and/or mobile internet device, according to one embodiment of the present invention.
[0023] FIG. **3**G**1** illustrates a healthcare (as a virtual doctor) related application of the social wallet electronic module, according to one embodiment of the present invention.
[0024] FIG. **3**G**2** illustrates a healthcare (as a virtual doctor) related application of the social wallet electronic module, according to another embodiment of the present invention, utilizing one or more machine learning algorithms (including an evolutionary algorithm), a Super System on Chip, a photonic neural learning processor (PNLP)/quantum computer (QC) and a blockchain.
[0025] FIG. **3**H**1** illustrates a consumer related application (e.g., a movie rental) of the social wallet electronic module, according to one embodiment of the present invention.
[0026] FIG. **3**H**2** illustrates a consumer related application (e.g., a movie rental) of the social wallet

electronic module, according to another embodiment of the present invention, utilizing one or more machine learning algorithms (including an evolutionary algorithm), a Super System on Chip, a photonic neural learning processor/quantum computer and a blockchain.
[0027] FIG. **3**I illustrates a consumer related application of user defined credit scoring application of the social wallet electronic module, according to one embodiment of the present invention, utilizing one or more machine algorithms (including an evolutionary algorithm), a Super System on Chip, a photonic neural learning processor/quantum computer and a blockchain.
[0028] FIG. **3**J illustrates a consumer related application of user defined service (e.g., a self-driving car rental) application of the social wallet electronic module, according to one embodiment of the present invention, utilizing one or more machine learning algorithms (including an evolutionary algorithm), a Super System on Chip, a photonic neural learning processor/quantum computer and a blockchain.
[0029] FIG. **3**K illustrates a consumer related application of user defined online/offline shopping application of the social wallet electronic module, according to one embodiment of the present invention, utilizing one or more machine learning algorithms (including an evolutionary algorithm), a Super System on Chip, a photonic neural learning processor/quantum computer, a blockchain and augmented reality.
[0030] FIG. **4** illustrates a block diagram of an object, according to one embodiment of the present invention.
[0031] FIG. **5**A illustrates a block diagram of the mobile internet device, according to one embodiment of the present invention.
[0032] FIG. **5**B illustrates a cross-section of a display component of the mobile internet device, according to one embodiment of the present invention.
[0033] FIG. **5**C illustrates a cross-section of a micro-electromechanical systems (MEMS) micro shutter, monolithically integrated with an array of thin-film transistors, according to one, embodiment of the present invention.
[0034] FIG. **5**D illustrates a cross-section of the display component integrated with a solar cell of the mobile internet device, according to one embodiment of the present invention.
[0035] FIGS. **5**E and **5**F illustrate cross-sections of the display component of the mobile internet device, according to one embodiment of the present invention.
[0036] FIG. **6** illustrates a block diagram of a sketch pad electronic module, according to one embodiment of the present invention.
[0037] FIG. **7** illustrates a block diagram of the sketchpad electronic module, according to one embodiment of the present invention.
[0038] FIG. **8**A illustrates a block diagram of a personal awareness assistant miniature electronic module, according to one embodiment of the present invention.
[0039] FIG. **8**B illustrates an application of the personal awareness assistant miniature electronic module, according to one embodiment of the present invention.
[0040] FIG. **9** illustrates a block diagram of a first system-on-chip for the mobile internet device, according to one embodiment of the present invention.
[0041] FIG. **10** illustrates a configuration of a nano-transistor, according to one embodiment of the present invention.
[0042] FIG. **11** illustrates a block diagram of a second system-on-chip (where a microprocessor is based on nano-transistors) for the mobile internet device, according to another embodiment of the present invention.
[0043] FIG. **12** illustrates a block diagram configuration of a memristor (or a phase change material based switching element), according to one embodiment of the present invention.
[0044] FIG. **13** illustrates a block diagram of a third system-on-chip, (where a microprocessor is based on memristors (or a phase change material based switching elements)) for the mobile internet device, according to another embodiment of the present invention.

[0045] FIG. **14** illustrates a block diagram configuration of a memristor (or a phase change material based switching element) with pre-neurons and post-neurons.
[0046] FIG. **15** illustrates a block diagram of a fourth system-on-chip (where a microprocessor is based on a neural network) for the mobile internet device, according to another embodiment of the present invention.
[0047] FIG. **16** illustrates a block diagram configuration of the memristors (or the phase change material based switching elements) with pre-neurons, post-neurons and nano-transistors, according to one embodiment of the present invention.
[0048] FIG. **17** illustrates a block diagram of a fifth system-on-chip (where a microprocessor is based on the neural network and nano-transistors) for the mobile internet device, according to another embodiment of the present invention.
[0049] FIGS. **18**A, **18**B and **18**C illustrate process steps for integrating one or two two-dimensional crystals on a semiconductor substrate, according to one embodiment of the present invention.
[0050] FIG. **19** illustrates a block diagram for attaching the above system-on-chip on a printed circuit board, according to one embodiment of the present invention.
[0051] FIGS. **20**A, **20**B and **20**C illustrate a block diagram of a cooling module for the above system-on-chip on the printed circuit board, according to one embodiment of the present invention.
[0052] FIGS. **21**A, **21**B and **21**C illustrate a block diagram of an interconnection between the above system-on-chips (via wavelength division multiplexing) on the printed circuit board, according to one embodiment of the present invention.
[0053] FIG. **21**D illustrates a cross section of a vertical cavity surface emitting laser, monolithically integrated with a modulator, according to one embodiment of the present invention.
[0054] FIGS. **22**A-**22**C illustrate three embodiments of a (optically enabled) Super System on Chip. It should be noted that FIGS. **22**A-**22**B are reproduced from U.S. Non-Provisional patent application Ser. No. 17/803,388 entitled "SUPER SYSTEM ON CHIP", filed on Jun. 15, 2022.
[0055] FIG. **22**D illustrates a block diagram of a (optically enabled) Super System on Chip in a two-dimensional (2-D) arrangement.
[0056] FIG. **23** illustrates an embodiment of a (personal) artificial intelligence (AI) based self-learning assistant in a block diagram.
[0057] FIGS. **24**A-**24**H illustrate an application of an intelligent (self-learning) subsystem and live view for remote shopping.
[0058] FIGS. **25**A-**25**E illustrate an application of an intelligent (self-learning) subsystem and live view for sharing a live game.

DETAILED DESCRIPTION OF THE INVENTION

[0059] FIG. **1**A illustrates an integrated application flow chart of a social wallet **100**. The social wallet **100** can be a social networking web portal and it can typically reside at a cloud based secure server, which can be coupled or integrated with a learning (including relearning) classical/quantum computer. The learning (including relearning) classical computer integrates one or more microprocessors, or one or more neural network based microprocessors, executing computer readable instructions and machine learning algorithm(s), stored on a non-transitory computer readable medium of the cloud based secure server to implement the social wallet **100**. The social wallet **100** can connect/access stored information from a data storage (preferably a cloud based secure data storage) component(s) **120**, can connect/access with information (preferably real time information) from an information source(s) (preferably real time information) **140**, can connect with a user(s) **160**, can connect with a merchant(s) **180**, can connect with a deposit account(s) **200**, can connect with a payment account(s) **220**, can connect with an object/array of objects **240**s, can connect with a node(s) **260**, can connect with a social wallet electronic module(s) **280**, can connect with a mobile internet appliance(s) **300**. Furthermore, the social wallet **100** can connect/access with a unified algorithm **320** and connect with the user **160** via one or more automated agents/software agents/bots (e.g., a chat bot/search bot)/multimodal bots, enhanced by the unified algorithm **320**.

The user **160** could type/talk with the social wallet **100** via the mobile internet appliance **300**: “Hi, should I renegotiate my car lease?” Instead of a search result, a lease bot enhanced by the unified algorithm **320** would pop up suggesting a better deal and close it for the user **160** in exchange for a small commission. Generally, a multimodal bot can include one or more computer implementable interfaces to comprehend (understand) and react to a voice, a text and a visual input (e.g., an image and/or a video).

[0060] It should be noted that the mobile internet appliance **300** can be a mobile wearable device, but the mobile wearable device is generally in a smaller form factor with respect to the mobile internet appliance **300**. Details of the mobile wearable device have been described/disclosed in U.S. Non-Provisional patent application Ser. No. 14/999,601 entitled “SYSTEM AND METHOD OF AMBIENT/PERVASIVE USER/HEALTHCARE EXPERIENCE”, filed on Jun. 1, 2016 (which claims benefit of priority to U.S. Provisional patent application No. 62/230,249 entitled “SYSTEM AND METHOD OF AMBIENT/PERVASIVE USER/HEALTHCARE EXPERIENCE”, filed on Jun. 1, 2015).

[0061] The unified algorithm **320** can consist of a software agent **340**, a fuzzy logic algorithm **360**, a predictive algorithm **380**, an intelligence rendering algorithm **400** and a self-learning (including relearning) algorithm **420**. It should be noted that the self-learning (including relearning) algorithm **420** can include a self-learning artificial intelligence algorithm and/or a self-learning neural network algorithm and/or an evolutionary algorithm, and/or a photonic neural learning processor enhanced machine learning algorithm and/or a quantum computer enhanced machine learning algorithm.

[0062] An evolutionary algorithm is an evolutionary computation in artificial intelligence. An evolutionary algorithm functions through the (similar to biological) selection process in which the least fit members of the population set are eliminated, whereas the fit members are allowed to survive and continue until better solutions are determined.

[0063] In other words, an evolutionary algorithm is a computer application which mimics the natural (biological) evolution in order to solve a complex problem. Over time, the successful members evolve to present the optimized solution to the complex problem. An evolutionary algorithm is a meta-algorithm, an algorithm for designing algorithms-eventually, the algorithms get pretty good at the task, based on three (3) pillars of innovation: [0064] 1. Meta-learning architectures (resulting in indirect coding), [0065] 2. Meta-learn learning algorithms (resulting in open ended search), [0066] 3. Generating effective learning environments (resulting in quality diversity).

[0067] The evolutionary algorithm is a class of stochastic search and optimization techniques obtained by natural selection and genetics. It is a population based algorithm by simulating the natural (biological) evolution. Individuals in a population compete and exchange information with one another.

[0068] There are three basic genetic operations: selection, crossover and random mutation.

[0069] For example, a procedure of an evolutionary algorithm is as follows: [0070] Step 1: Set t=0. [0071] Step 2: Randomize the initial population P(t). [0072] Step 3: Evaluate the fitness of each individual of P(t). [0073] Step 4: Select individuals as parents from P(t+1) based on the fitness. [0074] Step 5: Apply search operators (crossover and mutation) to parents, and generate P(t+1). [0075] Step 6: Set t=t+1. [0076] Step 7: Repeat step 3 to step 6 until the termination criterion is satisfied.

[0077] It should be noted that a conventional deep learning algorithm utilizes stochastic gradient descent (SGD), which improves an artificial neural network's over time by gradually reducing errors through an ongoing training with an existing dataset(s)—generally mapping inputs to outputs in known patterns over time, but it may not work properly in reinforcement learning (which is learning how to act/decide with only infrequent feedback signals or unknown outputs for given inputs without any pattern over time).

[0078] An artificial neuroevolution algorithm utilizes an evolutionary algorithm (similar to biological Darwinian evolution inspired by nature) with added safe/random mutations to grow/evolve/generate an artificial neural network's layers/rules/topology/parameters for better computing optimized outcomes/results.
[0079] Random mutations (may initially degrade an artificial neuroevolution algorithm before it improves) can allow evolving and reaching a decision toward achieving greater accuracy. Thus, an artificial neuroevolution algorithm can adapt dynamically and intelligently to unknown input signals.
[0080] Furthermore, an artificial neuroevolution algorithm can be coupled/connected with a cloud based expert system, as it requires significant computing power of a supercomputer.
[0081] Alternatively, an artificial neuroevolution algorithm can be coupled/connected with a quantum computer(s), as it is illustrated in FIGS. 64A-65C of U.S. Non-Provisional patent application Ser. No. 16/602,404 entitled "SYSTEM AND METHOD OF AMBIENT/PERVASIVE USER/HEALTHCARE EXPERIENCE", filed on Sep. 28, 2019. A quantum computer(s) has essentially an exponential computing power.
[0082] A photonic neural learning processor (can be useful for machine learning and/or image/pattern recognition and/or Big Data analysis) can be fabricated/constructed for example, by utilizing a cascaded configuration of interferometers (e.g., Mach-Zehnder type interferometers), 3-db couplers and waveguide based phase shifters. Heat applied to the waveguide base phase shifter(s) can direct light beams to change its shape. It should be noted that interferometer(s) and/or waveguide based phase shifter(s) can be fabricated/constructed, by utilizing a phase transition/phase change material for faster response to an external stimulus (e.g., heat or voltage) and/or integrated with saturable absorbers (e.g., graphene integrated saturable absorber). To reduce thermal cross-talk between the heating elements, thermal isolation trenches can be fabricated/constructed between the heating elements. Alternatively, the photonic neural learning processor can be fabricated/constructed for example as a network(s) of wavelength tunable/selective laser-integrated with an external modulator, when the external modulators are activated by an action of weighted electrical signals (from an array of memristors or by converting optical signals of distinct wavelengths from ring resonators/fast tunable ring resonators (e.g., fast tunable ring resonators incorporating phase transition material thin-film/quantum dot) based add/drop filters). The above network(s) can also utilize a network(s) of optical switches/fast optical switches. It should be noted that the photonic neural learning processor can be a standalone subsystem. Furthermore, various embodiments of system-on-chips/neural networks based system-on-chips have been described/disclosed described in U.S. Non-Provisional patent application Ser. No. 14/999,601 entitled "SYSTEM AND METHOD OF AMBIENT/PERVASIVE USER/HEALTHCARE EXPERIENCE", filed on Jun. 1, 2016 and the entire contents of this US Non-Provisional patent application are incorporated herein. Such a system-on-chip/neural networks based system-on-chip can replace the microprocessor/super-processor and enable cognitive/neural like computing. Such a system-on-chip/neural networks based system-on-chip can integrate the photonic neural learning processor via a network(s) of waveguide, thus enabling a hybrid electrical-photonic neural learning processor.
[0083] Alternatively, a photonic neural learning processor can be fabricated/constructed utilizing an array of optically induced phase transition material (e.g., vanadium dioxide (VO.sub.2)) based memristors.
[0084] At the heart of a quantum computer is a quantum bit (Qubit)—a basic unit of information analogous to a classical bit 0/1 represented by a transistor in a classical computer. The qubit is exponentially more powerful than the classical bit 0/1, because of its two unique properties: it can represent both 1 and 0 at the same time. But for qubit to be useful, it must achieve both quantum superposition (like being in two physical states simultaneously) and quantum entanglement (like what happens to one qubit can instantly affect the other qubit, even when they are physically

separated) and these two unique properties can be easily upset by a slightest disturbance (e.g., a material defect/vibration/fluctuating electric fields/noise). Therefore, qubits are extremely susceptible to error, without operating at an extremely low temperature.

[0085] However, operational qubits, by utilizing time crystals/implanted color centers (by utilizing silicon or germanium impurity ions) at a precise location on a diamond semiconductor/an array of (atomic force microscopy (AFM) manipulated) triangulene molecules at lower temperature (cooled by a quantum circuit refrigerator) can be possible.

[0086] Furthermore, a compact optical configuration can be realized by fabricating/constructing a network of silicon nitride waveguides on top of a substrate (e.g., glass/quartz). The network of silicon nitride waveguides can route light (e.g., light from quantum dot red/green/blue light emitting diodes/lasers or two-dimensional material based light sources). Above the silicon nitride waveguides, a layer (about 1 micron in thickness) of silicon dioxide thin-film or an electrically activated optically tunable material based thin-film can be fabricated/constructed. On top of the silicon dioxide thin-film or electrically activated optically tunable material based thin-film, there are transparent/indium tin oxide/niobium electrodes, integrated with tiny openings in the electrodes to allow light (which is guided via silicon nitride waveguides) to pass through. Beneath the tiny openings in the transparent/indium tin oxide/niobium electrodes, the waveguides in silicon nitride break into a series of sequential ridges to act as diffraction gratings in order to direct light down through the holes and concentrate the light into a beam narrow enough to activate/configure a spin of a color center in a semiconductor substrate.

[0087] Furthermore, integration of a surface normal light modulator (e.g., graphene based surface normal spatial light modulator (SLM)) with the diffraction gratings can also be realized.

[0088] Triangulene is similar to a fragment of graphene. It is made up of six hexagons of carbon joined along their edges to form a triangle, with hydrogen atoms around the sides.

[0089] A quantum computer enhanced machine learning algorithm is an approach that enables a quantum computer to learn/re-learn and to make predictions—by combining machine learning with quantum computation.

[0090] A quantum computer enhanced machine learning algorithm can be compiled on one or more microprocessors, or one or more neural network based microprocessors and downloaded onto the quantum computer-classical computer interface for execution. An application of the quantum computer enhanced machine learning algorithm can be hyper personalized advertising. But hyper personalized advertising can be further enhanced by utilizing the quantum computer enhanced machine learning algorithm (or its special case of deep learning algorithm) combined with topological data analysis (TDA).

[0091] For example, a machine learning algorithm (or a deep learning algorithm) combined with a topological data analysis (TDA) can enable: [0092] Creating an image of data, where every dot is an item or a group of similar items, [0093] Spotting patterns in data that would have been impossible to identify using a traditional algorithm(s), [0094] Segmenting in data on many levels, [0095] Analyzing any data (e.g., texts, images, sound and sensor data), [0096] Correlating more complex dependencies in data, and [0097] Learning even without supervision.

[0098] An example-application of the social wallet **100**, can allow the user **160** to get (a) his/her favorite coffee, as he/she approaches a Starbucks without placing an order, or (b) the object/array of objects **240***s* can notify the user **160** that his/her train is running late, or (c) the object/array of objects **240***s* can turn on room temperature, as the user **160** approaches his/her home, or (d) the user **160** can get a relevant coupon(s) (e.g., a movie coupon) with or without an augmented reality (AR) link, when the user **160** is engaging with an electronic display, where the electronic display is embedded with the object/array of objects **240***s* and/or near-field communication (NFC) tags and/or one-dimensional (1-D)/two-dimensional (2-D) quick response (QR) codes (e.g., a smart poster).

[0099] Furthermore, the social wallet **100** can connect with a location measurement component of the mobile internet appliance(s) **300**.

[0100] The social wallet **100** can act as an ultimate integrator (e.g., a Trusted Service Manager (TSM)) of many needs of the user **160**, connecting with other users **160***s* for various information and needs, transferring information between the other users **160***s*, securely transferring money to the deposit account **200** (e.g., a bank), securely transferring money to the payment account **220** (e.g., a bill payment account) and securely transferring money (e.g., a microloan) between the other users **160***s*.

[0101] The social wallet **100** can integrate a blockchain, instead of the Trusted Service Manager. A blockchain is a globally distributed ledger/database running on millions of devices and open to anyone, where not just information, but anything of value. In essence it is a shared, trusted public ledger that everyone can inspect, but which no single user controls. A blockchain creates a distributed document of (outputs/transactions) in the form of a digital ledger, which can be available on a network of computers/mobile internet devices **300***s*. When a transaction happens, the users **160***s* propose a record to the ledger. Records are bundled into blocks (groups for processing) and each block receives a unique fingerprint derived from the records it contains. Each block includes the fingerprint of the prior block, creating a robust and unbreakable chain. It is easy to verify the integrity of the entire chain and nearly impossible to falsify historic records. In summary, a blockchain is a public ledger of transactions, which critically provides trust, based upon mathematics rather than human relationships/institutions. A blockchain can replace the historically designed Trusted Service Manager.

[0102] Additionally, a blockchain can be integrated with the object/array of objects **240***s* and/or near-field communication tags and/or one-dimensional/two-dimensional quick response codes and such integration with a blockchain can build trust, reduce cost and accelerate smart (automated) transaction.

[0103] FIG. **1**B illustrates steps from **4001** to **4025**. In step **4001**, the social wallet **100** can connect/access stored information from the data storage (preferably the cloud based secure data storage) component(s) **120**. In step **4002**, the social wallet **100** can connect/access to the information source (preferably real time information) **140**.

[0104] In step **4003**, the social wallet **100** can connect to the user **160** via a profile. In step **4004**, the social wallet **100** can connect to the user **160** via online/offline message. In step **4005**, the social wallet **100** can connect to the user **160** via chat message. In step **4006**, the social wallet **100** can connect to the user **160** via broadcast message. In step **4007**, the social wallet **100** can connect to the user **160** via like/dislike vote. In step **4008**, the social wallet **100** can connect to the other users **160***s* for a collaborative purchase of a product and/or service (wherein the product and/or service can be coupled with a blockchain. For example, a car's parts, manufacturing, servicing, ownership and entire history can be coupled with the blockchain, creating trust between the user **160** and the merchant **180** and eliminating services provided by others).

[0105] In step **4009**, the social wallet **100** can connect to the merchant **180** via profile. In step **4010**, the social wallet **100** can connect to the merchant **180** via online/offline message. In step **4011**, the social wallet **100** can connect to the merchant **180** via chat message. In step **4012**, the social wallet **100** can connect to the merchant **180** via broadcast message. In step **4013**, the social wallet **100** can connect to the merchant **180** via bid. In step **4014**, the social wallet **100** can connect to the merchant **180** for price of the product or the service via bid in real time.

[0106] Furthermore, the bid/bid of the product or the service in real time can be based on game theory/algorithmic game theory/evolutionary game theory, wherein game theory/algorithmic game theory/evolutionary game theory can be coupled with fuzzy logic (or neuro-fuzzy logic) and/or Monte Carlo method and/or Markov method. In game theory, the users **160***s* and the merchants **180***s* become players, when their respective decisions coupled with the decisions made by other players, produce an outcome/output. The options available to players to bring about particular outcomes are called as strategies, which are linked to outcomes/outputs by a mathematical function that specifies the consequences of the various combinations of strategy choices by all players in a game. A coalition refers to the formation of sub-sets of players' options under coordinated strategies. In game theory, the core is the set of feasible allocations that cannot be improved upon by a coalition. An imputation X={x.sub.1, x.sub.2 . . . x.sub.n} is in the core of an n-person game if and only if for each subset, S of N:

[00001] $\text{.Math.}_{i=1}^{n} x_i \geq V(S)$

where V(s) is the characteristic function V of the subset S indicating the amount (reward) that the members of S can be sure of receiving, if they act together and form a coalition (or the amount of S can get without any help from players who are not in S). Above equation states that an imputation x is the core (that X is undominated), if and only if for every coalition S, the total of the received by the players in S (according to X) is at least as large a V(S). The core can also be defined by the equation below as the set of stable imputations:

[00002] $C: \{Y = (x_1, \text{.Math.}, x_n): \text{.Math.}_{i \in N} x_i = V(N) \text{and .Math.}_{i \in S} x_j \geq V(s) \forall S \subset N\}$

The imputation x is unstable through a coalition S, if the equation below is true, otherwise is stable.

[00003] $V(S) > \text{.Math.}_{i \in S} x_i$

The core can consist of many points. The size of the core can be taken as a measure of stability or how likely a negotiated agreement is prone to be upset. To determine the maximum penalty (cost) that a coalition in the network can be sure of receiving, the linear programming problem represented by the equation below can be used, when maximize x.sub.1+x.sub.2+x.sub.3+ . . . +x.sub.n

[00004] $\text{.Math.}_{i \in C} x_i \leq V(C) \forall C \subset N \text{subjectto} (x_{1,} x_2, \text{.Math.}, x_n) \geq 0$

Thus, as outlined above, a game theory based algorithm can account for any conflict in price negotiation. An example of the user **160** and N merchants **180***s* is described here. Using descending price [second price] for the product or service, the price starts high and continues to fall until one merchant **180** is left. Using ascending price [first price] for the product and/or service, the price starts low and continues to increase until one merchant **180** accepts. The user **160** can also suggest a price for the product and/or service to N merchants **180***s* in a time period and buy from the merchant **180**, who accepts the user **160**'s bid. The number of bidding rounds is determined by how frequently the user **160** can make an offer.

[0107] A set of instructions based fuzzy logic can be implemented as follows: (a) define linguistic variables and terms, (b) construct membership functions, (c) construct rule base, (d) convert crisp inputs into fuzzy values, utilizing membership functions (fuzzification), (e) evaluate rules in the rule base (inference), (f) combine the results of each rule (inference) and (g) convert outputs into non-fuzzy values (de-fuzzification).

[0108] Neuro-fuzzy logic integrates both artificial neural networks and fuzzy logic. Neuro-fuzzy logic can utilize fuzzy inference engine (with fuzzy rules for uncertainties) which is enhanced by artificial neural networks.

[0109] It should be noted that price can include brand value, resale value, warranty, delivery time, service contract, return policy and a value-added service(s).

[0110] Alternatively, the user can name his/her own price (wherein the price can include brand value, resale value, warranty, delivery time, service contract, return policy and a value-added service(s)) via conjoint analysis tool and relevant databases coupled with the social wallet **100**.

[0111] Alternatively, a back-and-forth negotiation until the user **160** and the merchant **180** reach a mutually agreeable price can be realized by a software (intelligent) agent, wherein the software agent is coupled with the social wallet **100**.

[0112] In step **4015**, the user **160** can deposit money (electronic scan of a money order and/or a check) and/or legally approved electronic cash (e.g., a bitcoin, digital gold currency and webmoney with traceable serial numbers) to the deposit account **200** via the social wallet **100**. In step **4016**, the user **160** can withdraw money from the deposit account **200** via the social wallet **100**.

[0113] In step **4017**, the user **160** can pay money to the payment account **220** via the social wallet **100**. In step **4018**, the user **160** can transfer/consolidate many payment accounts to the payment account **220** via the social wallet **100**.
[0114] In step **4019**, the user **160** can transfer money (e.g., a microloan) to the other users **160***s* via the social wallet **100**. In step **4020**, the social wallet **100** can communicate with the object **240**.
[0115] Furthermore, the objects **240***s*, near-field communication tags and/or one-dimensional/two-dimensional quick response codes can be embedded on an electronic display
[0116] Such communication with the objects **240***s* and/or near-field communication tags and/or one-dimensional/two-dimensional quick response codes can generate loyalty points in real time and can create personalized customer loyalty program when they are connected with the social wallet **100**.
[0117] In step **4021**, the social wallet **100** can communicate with the node **260**. Furthermore, the node **260** can communicate with the object/array of objects **240***s*.
[0118] In step **4022**, the social wallet **100** can communicate with the social wallet electronic module **280**.
[0119] In step **4023**, the social wallet **100** can communicate with the mobile internet device **300**.
[0120] In step **4024**, the social wallet **100** can communicate with the unified algorithm **320**. The unified algorithm **320** can consist of a software agent **340**, a fuzzy logic algorithm **360**, a predictive algorithm **380**, an intelligence rendering algorithm **400** and a self-learning (including relearning) algorithm **420**. In step **4025**, a software agent **340**, a fuzzy logic algorithm **360**, a predictive algorithm **380**, an intelligence rendering algorithm **400** and a self-learning (including relearning) algorithm **420** can communicate or couple with the said algorithms.
[0121] The intelligent rendering algorithm **400** can include algorithms such as: artificial intelligence, data interpretation, data mining, machine vision, natural language processing (NLP), neural network, pattern recognition and reasoning modeling. However, natural language processing emerges at the intersection of linguistics and artificial neural networks enabling machines the human-like ability to understand, interpret and generate natural language. Furthermore, the natural language processing can include either a transformer model or a diffusion model (which may be further augmented with an evolutionary based algorithm and/or a game theory based algorithm and/or Poisson flow generative model++ based algorithm).
[0122] Additionally, a neural network can approximate a function, but it is impossible to interpret the result in terms of a natural language. But an integration of the neural network and fuzzy logic in a neuro-fuzzy algorithm can provide both learning and readability. The neuro-fuzzy algorithm can use a fuzzy inference engine (with fuzzy rules) for modeling uncertainties, which is further enhanced through learning the various situations with a radial basis function. The radial basis function consists of an input layer, a hidden layer and an output layer with an activation function of hidden units. A normalized radial basis function with unequal widths and equal heights can be written as:

[00005] $${}_i(x)(\text{softmax}) = \frac{\exp(h_i)}{\sum_{i=1}^{n}\exp(h_i)} \quad h_i = \left(-\sum_{l=1}^{2}\frac{(X_i - u_{il})^2}{2\,{}_i^2}\right)$$

X is the input vector, uil is the center of the ith hidden node (i=1, . . . , 12) that is associated with the lth (l=1, 2) input vector, σi is a common width of the ith hidden node in the layer and softmax (hi) is the output vector of the ith hidden node. The radial basis activation function is the softmax activation function. First, the input data is used to determine the centers and the widths of the basis functions for each hidden node. Second, there is a procedure to find the output layer weights that minimizes a quadratic error between predicted values and target values. Mean square error can be defined as:

[00006] $$MSE = \frac{1}{N}\sum_{k=1}^{N}\left((\mathrm{TE})_k^{\exp} - (\mathrm{TE})_k^{\mathrm{cal}}1\right)^2$$

[0123] FIGS. **2**A, **2**B, **2**C, **2**D, **2**E, **2**F and **2**G illustrate a method flow chart of integration of the social wallet (the social networking web portal) **100** and electronic commerce in the following steps:

[0124] In step **4026**, the user **160** can log into the social wallet **100**. In step **4027**, the user **160** can set a privacy control in the social wallet **100**. In step **4028**, the user **160** can input his/her profile (e.g., gender, age group, income range, zip code, family members/friends' contacts) in the social wallet **100**.

[0125] In step **4029**, the unified algorithm **320** in the social wallet **100** can estimate the personal score of the user **160** by analyzing the profile, message history, chat history and data patterns (including purchase patterns). In step **4030**, the unified algorithm **320** in the social wallet **100** can set the personal score of the user **160**. The personal score of the user **160** can vary with time. In step **4031**, the social wallet **100** can record the personal score of the user **160** over time.

[0126] In step **4032**, the user **160** can authenticate in the social wallet **100**, by utilizing multi-level passwords and personalized questions. In step **4033**, the user **160** can authenticate in the social wallet **100**, by placing the social wallet electronic module **280**, at proximity to the near-field communication terminal (e.g., the near-field communication terminal of a computer/point-of-sale terminal) or by placing the mobile internet device **300**, at proximity to the near-field communication terminal (e.g., the near-field communication terminal of a computer/point-of-sale terminal), where the mobile internet device **300** also integrates the social wallet electronic module **280**. It should be noted that by placing the social wallet electronic module **280** at proximity to the near-field communication terminal of a computer, the social wallet **100**/social wallet electronic module **280**/user **160**'s profile/user **160** can be securely authenticated. Details of such secure authentication have been described/disclosed in U.S. Non-Provisional patent application Ser. No. 14/999,601 entitled METHOD “SYSTEM AND OF AMBIENT/PERVASIVE USER/HEALTHCARE EXPERIENCE”, filed on Jun. 1, 2016 (which claims benefit of priority to U.S. Provisional patent application No. 62/230,249 entitled “SYSTEM AND METHOD OF AMBIENT/PERVASIVE USER/HEALTHCARE EXPERIENCE”, filed on Jun. 1, 2015),

[0127] In step **4034**, the user **160** can also link the information about the product and/or service with or without an augmented reality link in the social wallet **100**.

[0128] Alternatively, in step **4035**, the user **160** can write a wanted ad for the product and/or service in the social wallet **100**.

[0129] In step **4036**, the unified algorithm **320** in the social wallet **100** can determine the location of the user **160** in real time by communicating with a location measurement component/miniature electronic module **1440** of the mobile internet device **300** of the user **160**.

[0130] In step **4037**, the unified algorithm **320** (in particular the software agent **340**) in the social wallet **100** can send out queries to the location specific merchants **180***s* for the product and/or service, wanted by the user **160**. If no offers from the location specific merchants **180***s* are found, then in step **4038**, the unified algorithm **320** can send out queries to the location specific merchants in an increment of some distance (e.g., 20 Km) from the current location of the user **160** in order to secure the product and/or service, wanted by the user **160**.

[0131] In step **4039**, the unified algorithm in the social wallet **100** can forward the offers (e.g., in the form of a text/e-mail link/picture mail/video mail) with or without an augmented reality link from the merchants **180***s* into the mobile internet device **300** of the user **160**, in real time (preferably via the user **160**'s profile in the social wallet **100**).

[0132] In step **4040**, the user **160** can optionally share the offers from the merchants **180***s* with the mobile internet devices **300***s* of the other users **160***s*, who are connected with the profile of the user **160**, in real time (preferably via the other users **160***s*′ profiles in the social wallet **100**).

[0133] In step **4041**, the other users **160***s*′ connected with the profile of the user **160** vote for like/dislike vote (like quorum sensing). In step **4042**, the user **160** can connect with the other users **160***s* for the collaborative purchase. In step **4043**, the unified algorithm **320** in the social wallet **100** can input the result of the like/dislike vote, in real time. In step **4044**, the unified algorithm **320** in the social wallet **100** can estimate a merchant score of the merchant **180** by analyzing many like/dislike votes.

[0134] Like (e.g., based on price, warranty, delivery time, service contract, return policy and a value-added service(s)) or dislike (e.g., based on price, warranty, delivery time, service contract, return policy and a value-added service(s)) votes can be computed by an algebraic equation (e.g., Fraction Likes=Number Like Votes/(Number Like Votes+Number Dislikes Votes). For example, if the Fraction Likes is below 0.2, then the merchant score is one star, if Fraction Likes is at least 0.2 but less than 0.4, then the merchant score is two stars and so on, if Fraction Likes is more than 0.8, then the merchant score is five stars.

[0135] Alternatively, the merchant score can be determined by (a) a statistical method (e.g., like/dislike vote distribution within segments of the users **160***s*) or a statistical method coupled with conjoint analysis (b) a quorum sensing clustering (QSC) algorithm or (c) a quorum sensing molecule (QSM) algorithm. [0136] Quorum sensing is a biological process, by which a community of bacteria interacts and coordinates with neighboring bacteria locally with no awareness of global information. In quorum sensing, bacteria do not communicate with each other directly, yet they influence and sense from a local environment, which assures local decision making optimization simultaneously. As an example of quorum sensing based decision making, assuming the number of users **160***s* is N in number and the user **160** has one of the two options (like (X) or dislike (Y)) with a constant probability r per time step. Probability r is independent of any actions taken by other users **160***s*. If the user **160** makes an option X, without the other users **160***s*, then the probability is aP.sub.x for option X and similarly, aP.sub.y for option Y. But, if the user **160** makes an option, wherein the other users **160***s* are present, then the probability is denoted as P.sub.x[a+(m−a)*{x.sup.k/(T.sup.k+x.sup.k)}], wherein “a” and “m” are minimum and maximum probability respectively for making an option. T is the quorum threshold at which the above probability is halfway between “a” and “m”. “k” determines the steepness of P.sub.x[a+(m−a)*{x.sup.k/(T.sup.k+x.sup.k)}]. If k is equal to 1, then the user **160** making an option is proportional to the number of the users **160***s* who made that option. If k is greater than 1, then P.sub.x[a+(m−a)*{x.sup.k/(T.sup.k+x.sup.k)}] can act as a step-like switch at the threshold T.

[0137] Quorum sensing enables a clustering algorithm. The quorum sensing clustering algorithm can define a cluster as a local region, where density is high and continually distributed. Thus, the quorum sensing clustering algorithm is capable of clustering datasets that are not linearly separable. By mimicking quorum sensing, the quorum sensing clustering algorithm can be decentralized, so that it is suitable for high speed parallel and distributed computing. Quorum sensing can enable a self-learning algorithm (inspired by a dynamic biological process), which can be immune to noise and outliers.

[0138] The quorum sensing molecule algorithm is at least based on non-linear dynamics observed at a population scale of a node.sub.i, calculating quorum sensing level/information sharing at the node; and then making use of the node.sub.i's quorum sensing level/information to calculate/influence the level/information at neighboring nodes such as, node.sub.j . . . node.sub.k.

[0139] The merchant score of the merchant **180** can vary over time. However, trust can be a critical element in influencing the user **160**'s decision making process. It represents the level of trust that the user **160** has toward recommending the merchant **180**. A trust based recommendation system (e.g., based on game theory) can be coupled with the social wallet **100**. The trust based recommendation system can incorporate: [0140] Analyzing similarity in knowledge and preferences between the users **160***s* in order to promote some liking and penalize some disliking, [0141] Analyzing recent liking or disliking of users **160***s* in active status, [0142] Evaluating the user **160***s* demographic information and/or habits.

[0143] In step **4045**, the social wallet **100** can record the merchant score. In step **4046**, the social wallet **100** can display the merchant score of the merchant **180**.

[0144] Furthermore, in step **4047**, if the estimated personal score of the user **160** exceeds a certain pre-determined value set by the social wallet **100**, then in step **4048**, the unified algorithm **320** (in particular the fuzzy logic algorithm **360**) in the social wallet **100** can determine other relevant products and/or services for the user **160**. In step **4049**, the social wallet **100** can send a coupon(s) (e.g., in the form of a text/e-mail link/picture mail/video mail) with or without an augmented reality link for the other relevant products and/or services from the merchants **180***s* to the mobile internet device **300** of the user **160**, in real time. In step **4050**, the user **160** can share the coupon(s) with the other users **160***s* by simply forwarding the coupon(s) to the other users **160***s*′ mobile internet devices **300***s*, in real time (preferably via the other users **160***s*′ profiles in the social wallet **100**).

[0145] In step **4051**, the user **160** can pay for the product and/or service via the social wallet **100**, or by the social wallet electronic module **280**, or by the mobile internet device **300**. The payment via the social wallet **100**, or by the social wallet electronic module **280**, or by the mobile internet device **300** can be coupled with a blockchain.

[0146] Furthermore, in step **4052**, the unified algorithm **320** (in particular the predictive algorithm **380**) in the social wallet **100** can initially determine a set of relevant users for a targeted advertisement for a specified product and/or service.

[0147] In step **4053**, the unified algorithm **320** in the social wallet **100** can send a coupon(s) (e.g., in the form of a text/e-mail link/picture mail/video mail) with or without an augmented reality link related to the specified product and/or service from the merchants **180***s* to the profiles of the above set of relevant users **160***s*.

[0148] In step **4054**, the above set of relevant users can share the coupon(s) (e.g., in the form of a text/e-mail link/picture mail/video mail) related to the specified product and/or service from the merchants **180***s* with the other users **160***s*′ mobile internet devices **300***s*, in real time (preferably via the other users **160***s*′ profiles in the social wallet **100**).

[0149] If a targeted advertisement campaign does not receive a response greater than at a pre-determined % (e.g., 10%), then in step **4055**, the unified algorithm **320** (in particular the predictive algorithm **380**, the intelligence rendering algorithm **400** and the self-learning (including relearning) algorithm **420**) in the social wallet **100** can iterate (fine-tune) to find another set of relevant users for the targeted advertisement, until the targeted advertisement would be concluded successful to stop, when the targeted advertisement campaign receives the response greater than at the pre-determined % (e.g., 10%).

Example A: Method Flow Chart from the User **160**'s Perspective

[0150] Any example in the above disclosed specifications is by way of an example only and not by way of any limitation (including following steps) [0151] (a): A computer implemented method incorporating: accessing, by the mobile internet device **300** or a mobile wearable internet device of the user **160**, via a wired or a wireless network, the social wallet **100** (e.g., a web portal) enabled by a learning or relearning classical computer, or a learning or relearning optical computer, or a learning or relearning quantum computer, wherein the social wallet **100** includes at least the user **160**'s profile, wherein the learning or relearning classical computer is one or more cloud computers, or premise computers, or mobile computers, wherein the learning or relearning classical computer includes one or more first microprocessors, or one or more first neural network based microprocessors, executing computer readable instructions and one or more machine learning algorithms (wherein the one machine learning algorithm includes an evolutionary algorithm), stored on a non-transitory computer readable medium to implement the social wallet **100**, wherein the learning or relearning optical computer includes one or more photonic neural learning processor, wherein the one photonic neural learning processor includes (i) optically induced phase transition material based memristors, or (ii) one or more optical waveguides, or one or more optical emitter systems, wherein the one optical emitter system is activated by weighted optical signals or weighted electrical signals, executing one or more machine learning algorithms to implement the social wallet **100**, wherein the learning or learning quantum computer includes one or more quantum bits (qubits), executing quantum computer enhanced algorithms or machine learning algorithms to implement the social wallet **100**, wherein the mobile internet device **300** or the mobile wearable internet device of the user **160** includes a wired connector or a wireless transceiver, wherein the mobile internet device **300** or the mobile wearable internet device of the user **160** further includes a second microprocessor, or a second neural network based microprocessor, wherein the mobile internet device **300** or the mobile wearable internet device of the user **160** is physically, or wirelessly communicatively coupled to the social wallet electronic module **280** of the user **160**, wherein the social wallet electronic module **280** of the user **160** electrically couples with the second microprocessor, or the second neural network based microprocessor of the mobile internet device **300** or the mobile wearable internet device of the user **160**, or the social wallet electronic module **280** of the user **160** includes a third microprocessor, or a microcontroller, wherein the social wallet electronic module **280** of the user **160** further includes a near-field communication component and a biometric sensor, wherein the mobile internet device **300** or the mobile wearable internet device of the user **160** accessing the social wallet **100**, wherein the said accessing the social wallet **100** includes obtaining a first biometric scan of the user **160** from the biometric sensor of the social wallet electronic module **280** of the user **160**, storing the first biometric scan of the user **160**, obtaining a second biometric scan of the user **160** from the biometric sensor of the social wallet electronic module **280** of the user **160**, wherein the second biometric scan of the user **160** is a current biometric scan of the user **160**, comparing the first biometric scan of the user **160** with the second biometric scan of the user **160** to authenticate the user **160**, or the social wallet electronic module **280** of the user **160**. [0152] (b): In response to at least (a), listing or linking, by the user **160**, a product or a service for purchase on the user **160**'s profile in the social wallet **100**, wherein the product or the service is linked with augmented reality. [0153] (c): In response to at least (a) and (b), automatically determining, by the social wallet **100**, that the user **160** is interested in purchasing the product or the service. [0154] (d): In response to at least (a), (b) and (c), automatically determining, by the social wallet **100**, a near real time location of the user **160**, wherein the near real time location of the user **160** is detected by a location measurement module of the mobile internet device **300** or the mobile wearable internet device of the user **160**, wherein the location measurement module may be selected from a group of a radio frequency identification module, a Bluetooth module, a WiFi module, a global positioning system module and indoor positioning system (IPS) module. [0155] (e): In response to at least (a), (b), (c) and (d), automatically querying, by the social wallet **100**, queried merchants **180***s* (e.g., the sellers) offering to sell the product or the service. [0156] (f): In response to at least (a), (b), (c), (d) and (e), automatically selecting, by the social wallet **100**, one of the queried merchant **180**, as a selected merchant **180** of the product or the service to purchase the product or the service, based on a distance from the near real time location of the user **160**. [0157] (g): In response to at least (a), (b), (c), (d), (e) and (f), automatically connecting, by the social wallet **100**, the user **160** with the selected merchant **180**. [0158] (h): In response to at least (a), (b), (c), (d), (e), (f) and (g), automatically forwarding, by the social wallet **180**, to the mobile internet device **300** or the mobile wearable internet device of the user **160**, in near real time, one or more sale offers to purchase the product or the service, from the selected merchant **180**. [0159] (i): In response to at least (a), (b), (c), (d), (e), (f), (g) and (h), automatically accepting, by the social wallet **180**, like votes and dislike votes for the queried merchant from the users **160***s*. [0160] (j): In response to at least (a), (b), (c), (d), (e), (f), (g), (h) and (i), automatically determining, a number of the like votes and a number of the dislike votes, further automatically determining, the merchant score for the queried merchant, based on the number of the like votes and the number of the dislike votes by an algebraic equation or a statistical method or a statistical method coupled with conjoint analysis or an algorithm based on quorum sensing. [0161] (k): In response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i) and (j), automatically displaying, the merchant score for the queried merchant **180**. [0162] (l): In response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i), (j) and (k), automatically (optionally) sending, by the social wallet **100**, to the mobile internet device **300** or the mobile wearable internet device of the user **160**, a coupon(s) for purchasing an additional product or service from the queried merchant(s) **180**. [0163] (m): In response to at least in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i), (j), (k) and (l), the selected merchant of the product or the service negotiating a price for the product or the service with the user **160**, or the user **160** picking his or her own price. [0164] (l): In response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i), (j), (k), (l) and (m), automatically accepting, by the social wallet **100**, payment by the user **160** for the product or the service, using the social wallet **100**, or the user **160**'s profile, or the near-field communication component of the social wallet electronic module **280** of the user **160**, based on one or more forwarded sale offers from the queried merchant(s) **180**. [0165] It should be noted that the learning or relearning classical computer can be coupled with the learning or relearning optical computer and/or the learning or relearning quantum computer. However, the learning or relearning classical computer can be integrated with the learning or relearning optical computer on a wafer (substrate) level, utilizing optical waveguides on the wafer (substrate). [0166] Wherein the said method steps in (a), (b), (c), (d), (e), (f), (g), (h), (i), (j), (k), (l), (m) and (n) are, at least an ordered combination or in an ordered sequence. [0167] The above method further incorporating the social wallet **100**, which is receiving a first input data from one or more sensors or a second input data from the users **160***s*. [0168] The above method further incorporating a set of instructions based on topological data analysis (TDA). [0169] The above method further incorporating the social wallet **100**, which is coupled with one or more automated agents/software agents/bots/multimodal bots. [0170] The above method further incorporating the merchant score, wherein the merchant score is coupled with a trust based recommendation system. [0171] The above method further incorporating the product or the service, wherein the product or the service or a service contract is coupled with a blockchain, wherein the block chain may be coupled with the social wallet **100**. [0172] The above method further incorporating the social wallet **100**, wherein the social wallet **100** is receiving a third input data from the near-field communication tag, or the quick response code, or the object(s) **240**. [0173] The above method further incorporating the social wallet **100**, wherein the user **160** is paying for the product or the service by transferring a currency or a check between the users **160***s* or between the user(s) **160***s* and the merchant(s) **180***s*.

[0174] The above method flow chart can be incorporated as a functional system.

Example B: Method Flow Chart from the Merchant **180**'s Perspective

[0175] Any example in the above disclosed specifications is by way of an example only and not by way of any limitation (including following steps)

[0176] The above method flow chart can be reversed, when the merchant **180** is a seller, when he/she is listing/linking the product or service to sell and looking for the users **160***s* (the buyers) in the increment of location distance with respect to the location distance of the merchant **180**. [0177] (a): A computer implemented method incorporating: accessing, by the mobile internet device **300** or a mobile wearable internet device of the user **160**, via a wired or a wireless network, the social wallet **100** (e.g., a web portal) enabled by a learning or relearning classical computer, or a learning or relearning optical computer, or a learning or relearning quantum computer, wherein the social wallet **100** includes at least the user **160**'s profile, wherein the learning or relearning classical computer is one or more cloud computers, or premise computers, or mobile computers, wherein the learning or relearning classical computer includes one or more first microprocessors, or one or more first neural network based microprocessors, executing computer readable instructions and one or more machine learning algorithms (wherein the one machine learning algorithm includes an evolutionary algorithm), stored on a non-transitory computer readable medium to implement the social wallet **100**, wherein the learning or relearning optical computer includes one or more photonic neural learning processor, wherein the one photonic neural learning processor includes (i) optically induced phase transition material based memristors, or (ii) one or more optical waveguides, or one or more optical emitter systems, wherein the one optical emitter system is activated by weighted optical signals or weighted electrical signals, executing one or more machine learning algorithms to implement the social wallet **100**, wherein the learning or learning quantum computer includes one or more quantum bits (qubits), executing quantum computer enhanced algorithms or machine learning algorithms to implement the social wallet **100**, wherein the mobile internet device **300** or the mobile wearable internet device of the user **160** includes a wired connector or a wireless transceiver, wherein the mobile internet device **300** or the mobile wearable internet device of the user **160** further includes a second microprocessor, or a second neural network based microprocessor, wherein the mobile internet device **300** or the mobile wearable internet device of the user **160** is physically, or wirelessly communicatively coupled to the social wallet electronic module **280** of the user **160**, wherein the social wallet electronic module **280** of the user **160** electrically couples with the second microprocessor, or the second neural network based microprocessor of the mobile internet device **300** or the mobile wearable internet device of the user **160**, or the social wallet electronic module **280** of the user **160** includes a third microprocessor, or a microcontroller, wherein the social wallet electronic module **280** of the user **160** further includes a near-field communication component and a biometric sensor, wherein the mobile internet device **300** or the mobile wearable internet device of the user **160** accessing the social wallet **100**, wherein the said accessing the social wallet **100** includes obtaining a first biometric scan of the user **160** from the biometric sensor of the social wallet electronic module **280** of the user **160**, storing the first biometric scan of the user **160**, obtaining a second biometric scan of the user **160** from the biometric sensor of the social wallet electronic module **280** of the user **160**, wherein the second biometric scan of the user **160** is a current biometric scan of the user **160**, comparing the first biometric scan of the user **160** with the second biometric scan of the user **160** to authenticate the user **160**, or the social wallet electronic module **280** of the user **160**. [0178] (b): In response to at least (a), listing or linking, by the merchant **180**, a product or a service for sale on the merchant's **180**'s profile in the social wallet **100**, wherein the product or the service is linked with augmented reality. [0179] (c): In response to at least (a) and (b), automatically determining, by the social wallet **100**, that the users **160***s*, who may be interested in purchasing the product or the service, listed or linked by the merchant **180** on the merchant's **180**'s profile in the social wallet **100**. [0180] (d): In response to at least (a), (b) and (c), automatically determining, by the social wallet **100**, a near real time location of the (interested) users **160***s*, wherein the near real time locations of the (interested) users **160***s* are detected by location measurement modules of the mobile internet devices **300***s* or the mobile wearable internet devices of the (interested) users **160***s*, wherein the location measurement module may be selected from a group of a radio frequency identification module, a Bluetooth module, a WiFi module, a global positioning system module and indoor positioning system module. [0181] (e): In response to at least (a), (b), (c) and (d), automatically connecting, by the social wallet **100**, the (interested) users **160***s* with the merchant **180**. [0182] (f): In response to at least (a), (b), (c), (d) and (e), automatically accepting, by the social wallet **180**, like votes and dislike votes for the merchant from the users **160***s*. [0183] (g): In response to at least (a), (b), (c), (d), (e) and (f), automatically determining, a number of the like votes and a number of the dislike votes, further automatically determining, the merchant score for the merchant, based on the number of the like votes and the number of the dislike votes by an algebraic equation or a statistical method or a statistical method coupled with conjoint analysis or an algorithm based on quorum sensing. [0184] (h): In response to at least (a), (b), (c), (d), (e), (f) and (g), automatically displaying, the merchant score for the queried merchant **180**. [0185] (i): In response to at least (a), (b), (c), (d), (e), (f), (g) and (h), automatically (optionally) sending, by the social wallet **100**, to the mobile internet devices **300***s* or the mobile wearable internet devices of the users **160***s*, a coupon(s) for purchasing an additional product or service from queried merchant(s) **180**. [0186] (j): In response to at least in response to at least (a), (b), (c), (d), (e), (f), (g), (h) and (i), the merchant of the product or the service negotiating a price for the product or the service with the users **160***s*, or the users **160***s* picking their own prices. [0187] (k): In response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i) and (j), automatically accepting, by the social wallet **100**, payment by

the user **160** for the product or the service, using the social wallet **100**, or the user **160**'s profile, or the near-field communication component of the social wallet electronic module **280** of the user **160**, based on one or more forwarded sale offers from the merchant **180**. [0188] Wherein the said method steps in (a), (b), (c), (d), (e), (f), (g), (h), (i), (j) and (k), are, at least an ordered combination or in an ordered sequence. [0189] The above method further incorporating the social wallet **100**, which is receiving a first input data from one or more sensors or a second input data from the users **160***s*. [0190] The above method further incorporating a set of instructions based on topological data analysis (TDA). [0191] The above method further incorporating the social wallet **100**, which is coupled with one or more automated agents/software agents/bots/multimodal bots. [0192] The above method further incorporating the merchant score, wherein the merchant score is coupled with a trust based recommendation system. [0193] The above method further incorporating the product or the service, wherein the product or the service or a service contract is coupled with a blockchain, wherein the block chain may be coupled with the social wallet **100**. [0194] The above method further incorporating the social wallet **100**, wherein the social wallet **100** is receiving a third input data from the near-field communication tag, or the quick response code, or the object(s) **240**.
[0195] The above method further incorporating the social wallet **100**, wherein the user **160** is paying for the product or the service by transferring a currency or a check between the users **160***s* or between the user(s) **160***s* and the merchant(s) **180***s*.
[0196] The above method flow chart can be incorporated as a functional system.
[0197] FIG. **2**G illustrates how the social wallet electronic module **280** and/or mobile internet device **300** can communicate symmetrically with the social wallet **100**, by utilizing a controller, a user interface layer, a user profile management layer, an encryption management layer, a protocol management layer and a communication management layer.
[0198] The Trusted Service Manager can consolidate/integrate/simplify various services with service providers (e.g., banks, phone companies and service providers).
[0199] FIG. **2**H illustrates how the social wallet **100** can function as the Trusted Service Manager to enable social electronic commerce, by utilizing the social wallet electronic module **280** and/or the mobile internet device **300**. The Trusted Service Manager can be coupled with various service providers such as banks, phone companies and service providers.
[0200] FIG. **3**A illustrates a block diagram of the social wallet electronic module **280** (preferably in a small form factor e.g., a SD/mini SD).
[0201] An external universal serial bus port **440** can connect with a universal serial bus (USB) connector **460**. The universal serial bus connector **460** can be electrically coupled with a universal serial bus interface **480**. The universal serial bus interface **480** can be electrically coupled with a computer readable medium (CRM) interface **500**.
[0202] The computer readable medium interface **500** can be electrically coupled with a solid state non-volatile (e.g., a flash/memristor based ReRAM) storage/memory **520** to store information. The solid state non-volatile storage/memory **520** can be partitioned to have both a private password protected storage/memory section (**520**-A) and a publicly viewable storage/memory section (**520**-B).
[0203] Furthermore, the solid state non-volatile memory **520** can store legally approved electronic cash (e.g., a bitcoin, digital gold currency and webmoney with traceable serial numbers).
[0204] The universal serial bus interface **480** and computer readable medium interface **500** can be electrically coupled with a microcontroller **540**.
[0205] A biometric (e.g., a fingerprint/retinal scan) sensor miniature electronic module **560** (an interface **580** and a component **600**) can be electrically coupled with the microcontroller **540**. The biometric sensor miniature electronic module **560** can enhance the security of the social wallet electronic module **280** or the user **160** by matching the stored biometric scan and an instant biometric scan at a point of presence or at a point of use for both online (Internet purchase) and offline (retail purchase) applications.

[0206] An advanced fingerprint sensor module can be fabricated/constructed by combining a silica colloidal crystal with a rubber, wherein the silica colloidal crystal can be dissolved in dilute hydrofluoric (HF) acid, leaving air voids in the rubber and thus creating an elastic photonic crystal. The elastic photonic crystal emits an intrinsic color, displaying three-dimensional (3-D) shapes of ridges, valley and pores of a fingerprint, when a finger is pressed onto the advanced fingerprint sensor. Details on the advanced fingerprint sensor have been described/disclosed in U.S. Non-Provisional patent application Ser. No. 11/952,001, entitled “DYNAMIC INTELLIGENT BIDIRECTIONAL OPTICAL AND WIRELESS ACCESS COMMUNICATION SYSTEM”, filed on Dec. 6, 2007 (now U.S. Pat. No. 8,073,331, issued on Dec. 6, 2011).
[0207] A near-field communication miniature electronic module **620** (an interface **640** and a component **660**) can be electrically coupled with the microcontroller **540**. Near-field communication is proximity range 13.56 MHz wireless (or radio) protocol.
[0208] Near-field communication has two key components: an initiator and a target. The initiator actively generates a radio frequency (RF) field that can electrically power a passive target without a battery.
[0209] A near-field communication tag contains simple data to perform a task (e.g., paying for the product and/or service and exchanging data between users). The near-field communication tag can securely store data (e.g., a personal identification number, debit/credit card information, loyalty card information, health record, physical access information, logical access information and digital rights access for local digital rights storage). But the near-field communication tag can also be re-writeable and be used as proximity based wireless charger.
[0210] A Wibree (a low electrical power, short-range wireless (or radio) protocol) miniature electronic module **680** (an interface **700** and a component **720**) can be electrically coupled with the microcontroller **540**.
[0211] A DASH7 (a low electrical power, -moderate-range wireless (or radio) protocol) miniature electronic module **740** (an interface **760** and a component **780**) can be electrically coupled with the microcontroller **540**. DASH7's electrical power requirements are about 10% of its next closest competitor (IEEE 802.15.4) and an even smaller fraction of WiFi and Bluetooth. With DASH7 miniature electronic module **720**, the user **160** passing by a restaurant at a low velocity (e.g., about 5 mph) could simply click a “get info” button to seek a customer review (or the merchant score) of the restaurant, before the user **160** decides to eat at the restaurant or not.
[0212] A Bluetooth miniature electronic module **800** (an interface **820** and a component **840**) can be electrically coupled with the microcontroller **540** to transmit and/or receive data.
[0213] A WiFi miniature electronic module **860** (an interface **880** and a component **900**) can be electrically coupled with the microcontroller **540** to transmit and/or receive data.
[0214] An ultra wideband miniature electronic module **920** (an interface **940** and a component **960**) can be electrically coupled with the microcontroller **540** to transmit and/or receive a vast quantity of data (e.g., a movie) in a short period of time.
[0215] A 60 GHz millimeter wave miniature electronic module **980** (an interface **1000** and a component **1020**) can be electrically coupled with the microcontroller **540** to transmit and/or receive a vast quantity of data (e.g., a movie) in a short period of time. The 60 GHz millimeter wave miniature electronic module **980** can enable applications such as (a) wireless docking and (b) distributed storage.
[0216] A software-defined radio **1040** can be fabricated/constructed by integrating a tunable antenna **1060**, a carbon nanotube tunable filter **1080** and an analog to digital converter **1100**.
[0217] The tunable antenna **1060** can tune in between 2 GHz and 3 GHz by utilizing a carbon nanotube. The tunable antenna **1060** can merge/integrate many antennas into one single antenna.
[0218] The software-defined radio **1040** and tunable antenna **1060** can be electrically coupled with the microcontroller **540**.
[0219] Additionally, a sensor (e.g., a wireless sensor-radio frequency identification (RFID)) **1120**

can be electrically coupled with the microcontroller **540**.

[0220] Furthermore, a line-of-sight optical transceiver **1140** (integrating an array of multi-color light source modulators **1160**, an array of photodiodes **1180**, two (2) waveguide combiner/decombiner **1200** and two (2) lenses **1220**) can be electrically coupled with the microcontroller **540**. The optical transceiver **1140** can transmit and/or receive a vast quantity of data (e.g., a movie) in a short period of time.

[0221] Additionally, an electrical power provider component (a thick-film/thin-film battery/solar cell/micro fuel-cell/supercapacitor) **1240** can be electrically coupled with the microcontroller **540**. It should be noted that the electrical power provider component **1240** can be enabled for wireless (e.g., near-field communication based) charging.

[0222] Furthermore, the microcontroller **540** can be replaced by a high-performance microprocessor **1360**.

[0223] Furthermore, the sensor **1120** can be a biosensor, wherein two embodiments of the biosensor are described in FIGS. **3**B and **3**C along with the Table-1 and Table-2:

TABLE-US-00001 TABLE 1 FIG. 3B Legend Description 1120-A Silicon Nanowire 1120-A1 Lipid Layer 1120-B Insulator (e.g., Silicon Dioxide) 1120-C Gate (e.g., Silicon Substrate) 1120-D Source Metal 1120-E Drain Metal 1120-F Receptor (e.g., Antibody Or Aptamer) 1120-G Biomarker Protein 1120-H Nanobattery 1120-I Nanotransmitter 1120-J Nanotube (e.g., Carbon Nanotube) 1120-K Negative Electrode 1120-L Positive Electrode

TABLE-US-00002 TABLE 2 FIG. 3C Legend Description 1120-A2 Chitosan 1120-B Insulator (e.g., Silicon Dioxide) 1120-C Gate (e.g., Silicon Substrate) 1120-D Source Metal 1120-E Drain Metal 1120-F Receptor (e.g., Antibody Or Aptamer) 1120-G Biomarker Protein 1120-H Nanobattery 1120-I Nanotransmitter 1120-J Nanotube (e.g., Carbon Nanotube) 1120-K Negative Electrode 1120-L Positive Electrode

[0224] FIG. **3**B illustrates an application of the sensor **1120**, as the biosensor, which can be integrated with the social wallet electronic module **280**. In the sensor **1120**, a silicon nanowire field effect transistor (FET), a source is identified by S, a drain is identified by D and a gate is identified by G. Furthermore, the silicon nanowire can be coated with a lipid layer and integrated with receptors on the lipid layer. The receptors can chemically bind with a biomarker protein (e.g., a disease biomarker protein)—thus giving rise to electrical signals (due to changes in the electrical properties of the silicon nanowire), further transmitted by a nanotube (e.g., a carbon nanotube) based wireless (or radio) transmitter. The nanotube based wireless (or radio) transmitter can be electrically powered with a nanobattery.

[0225] FIG. **3**C illustrates a disease detection application of the sensor **1120**, as the biosensor, which can be integrated with the social wallet electronic module **280**. In **1120**, chitosan proton (ionic) field effect transistor (H.sup.+ FET), a source is identified by S, a drain is identified by D and a gate is identified by G. Furthermore, chitosan can be integrated with receptors. The receptors can chemically bind with a biomarker protein (e.g., a disease biomarker protein)—thus giving rise to electrical signals (due to changes in the electrical properties of chitosan), further transmitted by a nanotube (e.g., a carbon nanotube) based wireless (or radio) transmitter. The nanotube based wireless (or radio) transmitter can be electrically powered with the nanobattery.

[0226] Furthermore, the sensor **1120**, as the biosensor can be integrated with the near-field communication miniature electronic module **620** on a human body to enable a smart biosensor to transmit vital health data to a near-field communication terminal. Alternatively, at least the sensor **1120**, as the biosensor can be implanted within a human body by utilizing a biocompatible outer case.

[0227] FIG. **3**D illustrates a solar cell **1240**-A as an electrical power provider component. About 2 microns thick meso-porous TiO.sub.2 thin-film **1240**-D can be coated with nanocrystals/nanoshells **1240**-E. The nanocrystals/nanoshells **1240**-E can cage/encapsulate light-absorbing organic dye molecules (e.g., porphyrins and/or phthalocyanines) **1240**-F. Furthermore, the nanocrystals/nanoshells **1240**-E can contain another specific molecule **1240**-G for energy transfer upon excitation.

[0228] The nanocrystals/nanoshells **1240**-E can also be varied in diameter to have an absorption over wider wavelength range in order for the solar cell **1240**-A to be more efficient (for light to electricity conversion).

[0229] Furthermore, the solar cell **1240**-A could be made more efficient (for light to electricity conversion) with an addition of an array of nanotubes (e.g., carbon or boron nitride nanotubes) **1240**-H.

[0230] The meso-porous TiO.sub.2 thin-film **1240**-D can be sandwiched between two electrodes: indium tin oxide transparent front electrode **1240**-J and back metal (e.g., aluminum, silver or platinum) electrode **1240**-C.

[0231] Furthermore, the back metal electrode **1240**-C can be fabricated/constructed with nanocorrugated plasmonic reflectors to trap more residual light inside the solar cell **1240**-A.

[0232] The meso-porous TiO.sub.2 thin-film **1240**-D can be immersed within a liquid ionic electrolyte solution **1240**-I.

[0233] FIG. **3**E illustrates the solar cell **1240**-A as the electrical power provider component. Triple junction semiconductor epitaxial layers **1240**-P can be purchased from Microlink Devices. The critical element of this embodiment is (a) chemically separating (by selectively etching 50 nanometers thick AlAs layer in hydrofluoric acid), (b) lifting (by covering the patterned front device side with a black wax) triple junction semiconductor epitaxial layers **1240**-P, all other relevant layers (such as GaAs layer **1240**-M and InGaAlP layer **1240**-N), p-metallization **1240**-R and n-metallization **1240**-S, (c) bonding onto the glass substrate **1240**-B with an adhesion layer **1240**-L and (d) finally dissolving the black wax in trichloroethylene (TCE).

[0234] FIG. **3**F illustrates a cross-section of the solar cell **1240**-A as the electrical power provider component. The critical element of this embodiment is (a) chemically separating (by selectively etching 50 nanometers thick AlAs layer in hydrofluoric acid), (b) lifting (by covering the patterned front device side with a black wax) three-dimensional quantum dot superlattice of InAs **1240**-Q, all other relevant layers (such as GaAs layer **1240**-M and InGaAlP layer **1240**-N), p-metallization **1240**-R and n-metallization **1240**-S, (c) bonding onto the glass substrate **1240**-B with an adhesion layer **1240**-L and (d) finally dissolving the black wax in trichloroethylene.

[0235] FIG. **3**G**1** illustrates a healthcare related application of the social wallet electronic module **280**: how the social wallet electronic module **280** can be utilized to obtain healthcare related advice from a healthcare expert system (a virtual doctor) at a cloud server. The social wallet electronic module **280** can be integrated with the sensor **1120** as the biosensor or a point-of-care diagnostic device. In step **4056**, the social wallet electronic module **280** transmits wireless (or radio) network settings to the cloud based healthcare expert system (the virtual doctor) **100**A. In step **4057**, the social wallet electronic module **280** establishes wireless (or radio) connection with the cloud based healthcare expert system (the virtual doctor) **100**A. In step **4058**, the social wallet electronic module **280** establishes security verification with the cloud based healthcare expert system (the virtual doctor) **100**A. In step **4059**, the social wallet electronic module **280** transfers the user's health related data to the cloud based healthcare expert system (the virtual doctor) **100**A. In step **4060**, the social wallet electronic module **280** receives expert healthcare advice from the cloud based healthcare expert system (the virtual doctor) **100**A. In step **4061**, the user **160** pays using the social wallet electronic module **280** (integrated with a biosensor) for the expert advice received from the cloud based healthcare expert system (the virtual doctor) **100**A. Various embodiments of the point-of-care diagnostic device have been described/disclosed in U.S. Non-Provisional patent application Ser. No. 14/999,601 entitled “SYSTEM AND METHOD OF AMBIENT/PERVASIVE USER/HEALTHCARE EXPERIENCE”, filed on Jun. 1, 2016 (which claims benefit of priority to U.S. Provisional patent application No. 62/230,249 entitled “SYSTEM AND METHOD OF AMBIENT/PERVASIVE USER/HEALTHCARE EXPERIENCE”, filed on Jun. 1, 2015), U.S.

[0236] Non-Provisional patent application Ser. No. 14/120,835 entitled "AUGMENTED REALITY PERSONAL ASSISTANT APPARATUS", filed on Jul. 1, 2014 and U.S. Non-Provisional patent application Ser. No. 13/663,376 entitled "OPTICAL BIOMODULE FOR DETECTION OF DISEASES", filed on Oct. 29, 2012 (or "OPTICAL BIOMODULE FOR DETECTION OF DISEASES", U.S. Pat. No. 9,557,271, issued on Jan. 31, 2017).

[0237] FIG. **3**G**2** is similar to FIG. **3**G**1**, except it illustrates a healthcare (as a virtual doctor) related application of the social wallet electronic module, according to another embodiment of the present invention, utilizing one or more machine learning algorithms (including an evolutionary algorithm) represented by "Fazila", a Super System on Chip, a photonic neural learning processor/quantum computer and a blockchain. The cloud based healthcare expert system is coupled with the social wallet electronic module **280** (integrated with a biosensor) via a blockchain.

[0238] The cloud based healthcare expert system is coupled with "Fazila" (an integrated software module for machine learning). Furthermore, "Fazila" can be coupled with a Super System on Chip and a photonic neural learning processor/quantum computer.

[0239] Compared to FIG. **3**G**1**, in FIG. **362**, the step **4058** is replaced by the step **4058**-A, wherein the social wallet electronic module **280** verifies a user identification via biometrical identification and/or near-field communication, the step **4059** is replaced by the step **4059**-A, wherein the social wallet electronic module **280** transfers confidential health data to the cloud based healthcare expert system (the virtual doctor) **100**A via a blockchain, the step **4060** is replaced by the step **4060**-A, wherein the social wallet electronic module **280** receives confidential healthcare advice from the cloud based healthcare expert system (the virtual doctor) **100**A via a blockchain, the step **4061** is replaced by the step **4061**-A, wherein the social wallet electronic module **280** pays for the confidential healthcare advice via a blockchain. It should be noted that the social wallet electronic module **280** can be replaced by a mobile internet device **300**.

[0240] Details of "Fazila", a Super System on Chip and a quantum computer have been described/disclosed (e.g., FIG. 10A of U.S. Non-Provisional patent application Ser. No. 16/602,404) in U.S. Non-Provisional patent application Ser. No. 16/602,404 entitled "SYSTEM AND METHOD OF AMBIENT/PERVASIVE USER/HEALTHCARE EXPERIENCE", filed on Sep. 28, 2019.

[0241] FIG. **3**H**1** illustrates a consumer related application of the social wallet electronic module **280**: how the social wallet electronic module **280** can be utilized to obtain a movie from the cloud based movie storage system **100**B. In step **4056**, the social wallet electronic module **280** transmits a wireless (or radio) network setting(s) to the cloud based movie storage system **100**B. In step **4057**, the social wallet electronic module **280** establishes a wireless (or radio) connection with the cloud based movie storage system **100**B. In step **4058**, the social wallet electronic module **280** establishes a security verification with the cloud based movie storage system **100**B. In step **4060**, the social wallet electronic module **280** receives (downloads) a movie from the cloud based movie storage system **100**B. In step **4061**, the user **160** pays for the movie received (downloaded) from the cloud based movie storage system **100**B using the social wallet electronic module **280** for a specified period of time, after the specified period of time, the received (downloaded) movie can be automatically disabled by a set of instructions (e.g., computer codes).

[0242] FIG. **3**H**2** is similar to FIG. **3**H**1**, except it illustrates a consumer related application (e.g., a movie rental) of the social wallet electronic module, according to another embodiment of the present invention, utilizing one or more machine learning algorithms (including an evolutionary algorithm), a Super System on Chip, a photonic neural learning processor/quantum computer and a blockchain.

[0243] Compared to FIG. **3**H**1**, in FIG. **3**H**2**, the step **4058** is replaced by the step **4058**-A, wherein the social wallet electronic module **280** verifies a user identification via biometrical identification and/or near-field communication, the step **4060** is replaced by the step **4060**-A, wherein the social wallet electronic module **280** receives a movie from the cloud based movie storage system **100**B via a blockchain, the step **4061** is replaced by the step **4061**-A, wherein the social wallet electronic module **280** pays for the movie via a blockchain. It should be noted that the social wallet electronic module **280** can be replaced by a mobile internet device **300**.

[0244] Alternatively, a movie storage system can be located at widely distributed (and conveniently located) kiosks, instead of the cloud based movie storage system **100**B. Furthermore, the rented movie can have coded expiry date, after that the rented movie will not run.

[0245] FIG. **3**I illustrates a consumer related application of user defined credit scoring application of the social wallet electronic module, according to one embodiment of the present invention, utilizing one or more machine algorithms (including an evolutionary algorithm), a Super System on Chip, a photonic neural learning processor/quantum computer and a blockchain.

[0246] Critically in FIG. **3**I, in step **4062**-A, wherein the social wallet electronic module **280** transmits confidential history/log of a user's bill payments to the cloud based credit scoring system **100**C via a blockchain, in step **4063**-A, the social wallet electronic module **280** receives a user's credit score from the cloud based credit scoring system **100**C.

[0247] FIG. **3**J illustrates a consumer related application of user defined service (e.g., a self-driving car rental) application of the social wallet electronic module, according to one embodiment of the present invention, utilizing one or more machine learning algorithms (including an evolutionary algorithm), a Super System on Chip, a photonic neural learning processor/quantum computer and a blockchain.

[0248] Critically in FIG. **3**J, in step **4064**-A, a user receives user specified service (e.g., a self-driving car rental) from a cloud based user service system **100**D.

[0249] FIG. **3**K illustrates a consumer related application of user defined online/offline shopping application of the social wallet electronic module, according to one embodiment of the present invention, utilizing one or more machine learning algorithms (including an evolutionary algorithm), a Super System on Chip, a photonic neural learning processor/quantum computer, a blockchain and augmented reality. Following FIGS. 9A-9D of U.S. Non-Provisional patent application Ser. No. 16/602,404 entitled "SYSTEM AND METHOD OF AMBIENT/PERVASIVE USER/HEALTHCARE EXPERIENCE", filed on Sep. 28, 2019, the social wallet electronic module can be replaced by a physical card, wherein the physical card can include a magnetic strip, a wireless communication chip (e.g., a near-field communication chip), a dynamically reconfigured/changed card verification value number. The physical card can also include a biometric sensor. A biometric sensor can be a fingerprint/face identification/voice sensor.

[0250] Critically in FIG. **3**K, in step **4065**-A, a user receives user specified product/service from the cloud based online/offline shopping system **100**E with or without an augmented reality link, in step **4065**-A, a user finds the user specified product/service in an offline (retail) store.

[0251] FIG. **4** illustrates a block diagram of the object **240**. The object **240** has ultra-low electrical power consumption and miniature medium performance microprocessor (e.g., an Ambiq Micro or InAs-on-Insulator based microprocessor or a memristor) **1260**, which can be electrically coupled with: (a) a sensor (e.g., a wireless sensor-a radio frequency identification) **1120**, (b) an optional IP/micro IP/light weight IP address **1280**, (c) a miniature memory/storage (e.g., a memristor) **1300**, (d) an embedded tiny operating algorithm/executable instructions **1320** (e.g., a Tiny OS), (e) a miniature low electrical power "object-specific" miniature wireless (or radio) transmitter (e.g., a radio frequency identification and/or a Wibree and/or a Bluetooth and/or a WiFi and/or a near-field communication) with a miniature antenna **1340** and (f) an "object-specific" electrical power provider component **1240**-T (e.g., the solar cell **1240**-A in a miniature form factor).

[0252] Furthermore, the object **240** can have an outer external case. The object **240** can also be a biological object on or within (e.g., implanted by utilizing the outer external case, which is biocompatible) a human body.

[0253] The object **240** can utilize semiconductor fabrication, micro-electromechanical systems fabrication, plastic electronics fabrication, printed electronics fabrication, multi-chip module fabrication (packaging), three-dimensional fabrication (packaging) and microfluidic fabrication.

[0254] The array of objects **240***s* can connect to the node (e.g., the node with an internet connection) **260**. The node **260** can map, sense, measure, collect, aggregate, compare information collected from the array of objects **240***s*. The node **260** can share/communicate information with the social wallet **100** and/or electronic social wallet electronic module **280** and/or mobile internet device **300**.

[0255] The objects **240***s* and node **260** can self-register with a blockchain for integration. For example, when the object **240**/node **260** detects something wrong, it can automatically request a smart repair contract. Additionally, the node **260** can be integrated with one or more automated agents/software agents/bots/multimodal bots.

[0256] Furthermore, the electronic social wallet electronic module **280** and/or mobile internet device **300** can proximity contact or physically contact with the object **240** to communicate for relevant information.

[0257] FIG. **5**A illustrates a block diagram of the mobile internet device **300**. It has the high performance microprocessor (e.g., Intel's x86 based Medfield) **1360**, which can be electrically coupled with (a) the social wallet electronic module **280** (in this case the social wallet electronic module **280** does not require the separate microcontroller **540** or the microprocessor **1360**, but it shares the high performance microprocessor **1360** of the mobile internet device), (b) a general data storage electronic module **520**-C, (c) the IP/micro IP/light weight IP address **1280**, (d) a lab-on-chip electronic module (a biological diagnostics electronic module) **1360**, (e) an embedded operating algorithm **1380** stored in a general data storage electronic module **520**-C, (f) an internet security algorithm (internet firewall/spyware/user-specified security control and authentication) **1400**, (g) a one-dimensional/two-dimensional barcode/quick response code reader **1420**, (h) a miniature wireless (or radio) electronic module (e.g., a radio frequency identification/Bluetooth/WiFi/global positioning system location (GPS) with an antenna(s)) **1440** for an indoor/outdoor location measurement, (i) an electronic compass **1460**, (j) two (2) cameras (a 180 degree rotating camera is preferred, instead of two cameras—one for video chat and one for photo taking) **1480**, (k) a video conferencing (integrated with dynamic video compression algorithm) system-on-chip **1500**, (l) a display component **1520**, (m) a micro-projector **1540**, (n) a sketch pad (with a write/erase option) electronic module **1560** with a stylus **1580**, (o) a communication wireless (or radio) transceiver electronic module (e.g., WiMax/LTE) with an antenna(s) **1600**, (p) a personal awareness assistant miniature electronic module **1620**, (q) a voice-to-text-to-voice conversion algorithm **1640** and (r) an algorithm **1660**

[0258] Furthermore, the mobile internet device **300** has the electrical power provider component. (e.g., battery/solar cell/micro fuel-cell/supercapacitor) **1240**. It should be noted that the electrical power provider component **1240** can be enabled for wireless (e.g., near-field communication based) charging.

[0259] A multi-touch high definition liquid crystal display (integrated with an array of thin-film transistors on indium gallium zinc oxide) can be utilized as the display component **1520**.

[0260] Organic light emitting (red, green and blue) diodes driven by an array of organic thin-film transistors on an organic substrate (e.g., plastic) can also be utilized as the rolled up/foldable/stretchable display component **1520**. The rolled up/foldable/stretchable display component **1520** can minimize its size related distinction between a portable computer (e.g., a laptop) and the mobile internet device **300**. Furthermore, a foldable display may develop micro-cracks, which can be reduced/prevented by a protective transparent layer that generally includes a polymer (host) material and ceramic particles (e.g., Al.sub.2O.sub.3, AlON or SiAlON particles).

[0261] Additionally, the rolled up/foldable/stretchable display component **1520** can be constructed from a graphene sheet and/or an organic light-emitting diode connecting/coupling/interacting with a printed organic transistor and a rubbery conductor (e.g., a mixture of a carbon nanotube/gold conductor and a rubbery polymer) with a touch/multi-touch sensor.

[0262] Furthermore, the display component **1520** can enable a dual-view to show entirely two separate scenes simultaneously.
[0263] Furthermore, organic light emitting diodes based display component **1520** may be more transparent to a short wavelength infrared (SWIR) photodetector than a near infrared (NIR) photodetector. Such a short wavelength infrared photodetector (two-dimensional (2-D)) array (e.g., 1280×1024 pixel format) can be fabricated utilizing InGaAs material or quantum dots or germanium (Ge) on silicon (Si) substrate or even GeSn/Ge MQW on silicon substrate. Furthermore, short wavelength infrared photodetector can include a resonant cavity enhanced (RCE) (e.g., utilizing a bottom Bragg reflecting mirror and a top Bragg reflecting mirror.
[0264] It should be noted that the short wavelength infrared photodetector (two-dimensional) array can be replaced by a (two-dimensional) array of phototransistors or a (two-dimensional) array of microscaled (e.g., about 0.5 microns to 3 microns diameter etched mesa) phototransistors. Microscaled mesa phototransistors may offer higher detection sensitivity.
[0265] Such a short wavelength infrared photodetector (two-dimensional) array can be electrically coupled with one or more electronic circuits (e.g., a read-out integrated circuit-ROIC) via fine pitch indium bump bonding or fine pitch metal-to-metal bonding.
[0266] Additionally, the read-out integrated circuit can also be electrically coupled with a Super System on Chip for ultrafast data processing, image processing/recognition, deep learning/meta-learning and self-learning. Unlike traditional central processing units capable of processing serialized instruction streams, neural processing units can be capable of processing machine learning workloads (which may be highly parallelizable with high regularity in the computational patterns of deep neural networks). It should be noted that in some applications, a Super System on Chip may be replaced by a neural processor (e.g., Google's Tensor Processing Unit-TPU).
[0267] Organic light emitting diodes based display component **1520** can be positioned above a short wavelength infrared photodetector (two-dimensional) array/phototransistor (two-dimensional) array/microphototransistor (two-dimensional) array (also a short wavelength infrared photodetector (two-dimensional) array/phototransistor (two-dimensional) array/microphototransistor (two-dimensional) array can include a read-out integrated circuit) for a three-dimensional (3-D) display coupled sensor (e.g., a under-display sensor) on the mobile internet device **300** (or on a computer).
[0268] For nighttime application, a three-dimensional display coupled sensor may require one or more eye-safe lasers (e.g., vertical cavity surface emitting laser (VCSEL) diodes/edge emitting laser diodes) operating in short wavelength infrared range and also compatible with a short wavelength infrared photodetector (two-dimensional) array/phototransistor (two-dimensional) array/microphototransistor (two-dimensional) array. A short wavelength infrared photodetector (two-dimensional) array/phototransistor (two-dimensional) array/microphototransistor (two-dimensional) array may also include one or more single photon avalanche diodes (SPADs). An edge emitting laser diode with higher output power (operating in short wavelength infrared range) may enable longer range and higher reliability of three-dimensional display coupled sensing.
[0269] Furthermore, (i) one or more laser diodes and (ii) a short wavelength infrared photodetector (two-dimensional) array/phototransistor (two-dimensional) array/microscaled phototransistor (two-dimensional) array can be integrated/co-placed on a suitable common substrate by heterogeneous fabrication process (e.g., micro-transfer printing (MTP)).
[0270] Furthermore, organic light emitting diodes based display component **1520** may be positioned above a short wavelength infrared photodetector (two-dimensional) array/phototransistor (two-dimensional) array/microphototransistor (two-dimensional) array (also a short wavelength infrared photodetector (two-dimensional) array/phototransistor (two-dimensional) array/microphototransistor (two-dimensional) array can include a read-out integrated circuit and a processor/Super System on Chip) for a three-dimensional (3-D) intelligent/smart display coupled sensor on the mobile internet device **300** (or on a computer).
[0271] The Super System on Chip that includes memristors can be electrically and/or optically

controlled. For example, memristors made of a phase transition material (e.g., vanadium dioxide-VO.sub.2) can be electrically and/or optically controlled. Alternatively memristors made of a phase change material (e.g., Ge.sub.2Sb.sub.2Te.sub.5 (GST)) can be either electrically or optically controlled.
[0272] There is a distinction between a phase transition material and a phase change material. For example, in case of a phase transition material, such as vanadium dioxide, there is volatile phase transition from insulating (about band gap 0.7 eV) monoclinic crystalline phase to metallic (i.e., bands gap collapse and delocalized electrons) in a rutile crystalline phase with a corresponding change in symmetry and reduced (by half) unit cell. But in case of a phase change material, such as GST, there is non-volatile phase change from glassy phase to crystalline phase with accompanying change of dielectric function.
[0273] A three-dimensional display coupled sensor or a three-dimensional intelligent/smart display coupled sensor may enable [0274] biometric sensing, [0275] range sensing, [0276] sensing in mixed reality/metaverse on the mobile internet device **300** (or on a computer). [0277] Furthermore it can identify, delineate, classify and render image content (or with avatars it can experience real time eye contact and facial expressions).
[0278] Details of various embodiments of a detector array have been described/disclosed in U.S. Non-Provisional patent application Ser. No. 16/602,404 entitled “SUPER SYSTEM ON CHIP”, filed on Sep. 28, 2019.
[0279] Details of various embodiments of a Super System on Chip have been described/disclosed in U.S. Non-Provisional patent application Ser. No. 16/602,404 entitled “SUPER SYSTEM ON CHIP”, filed on Sep. 28, 2019 and subsequently in U.S. Non-Provisional patent application Ser. No. 17/803,388 entitled “SUPER SYSTEM ON CHIP”, filed on Jun. 15, 2022.
[0280] If both the display component **1520** and electrical power provider component are thinner, then the mobile internet device **300** would be thinner. A thinner organic battery component can be fabricated/constructed as follows: an organic battery utilizes push-pull organic molecules, wherein after an electron transfer process, two positively charged molecules are formed, which are repelled by each other like magnets. By installing a molecular switch an electron transfer process can proceed in the opposite direction. Thus, forward and backward switching of an electron flow can form a basis of an ultra-thin, light weight and electrical power efficient organic battery. Details on the thinner organic battery have been described/disclosed in U.S. Non-Provisional patent application Ser. No. 11/952,001, entitled “DYNAMIC INTELLIGENT BIDIRECTIONAL OPTICAL AND WIRELESS ACCESS COMMUNICATION SYSTEM”, filed on Dec. 6, 2007 (now U.S. Pat. No. 8,073,331, issued on Dec. 6, 2011).
[0281] Furthermore, the algorithm **1660** includes: (a) a physical search algorithm, (b) an algorithm-as-a-service, (c) an intelligent rendering algorithm (e.g., artificial intelligence, behavior modeling, data interpretation, data mining, fuzzy logic, machine vision, natural language processing, neural network, pattern recognition and reasoning modeling) and (d) a self-learning (including relearning) algorithm. It should be noted that the self-learning (including relearning) algorithm can include a self-learning artificial intelligence algorithm and/or a self-learning neural network algorithm
[0282] In the context of the mobile internet device **300**, data can be compared with respect to a set of parameters to learn or relearn continuously by analyzing patterns of data, where patterns of data can consist/utilize/couple with the algorithm **1660**.
[0283] The algorithm **1660** of the mobile internet device **300** includes the intelligent rendering algorithm (e.g., artificial intelligence, behavior modeling, data interpretation, data mining, fuzzy logic, machine vision, natural language processing, neural network, pattern recognition and reasoning modeling). Additionally, the mobile internet device **300** can be integrated with one or more automated agents/software agents/bots (including a voice/text interface), enhanced by the algorithm **1660**.
[0284] In the context of the social wallet **100**, data can be compared with respect to a set of

parameters to learn or relearn continuously by analyzing patterns of data, where patterns of data can consist/utilize/couple with the fuzzy logic algorithm **360**, the intelligence rendering algorithm **400** and the self-learning (including relearning) algorithm **420**. It should be noted that the self-learning (including relearning) algorithm **420** can include a self-learning artificial intelligence algorithm and/or a self-learning neural network algorithm

[0285] Furthermore, this continually learned analysis along with the predictive algorithm **380** can enable the social wallet **100** to identify a set of users with particular parameters for a targeted advertisement.

[0286] The antenna for the communication wireless (or radio) transceiver **1600** of the mobile internet device **300** can be fabricated/constructed from metamaterial. Metamaterial is a material of designer crystal structure combining two materials (e.g., lead selenide and iron oxide).

[0287] Furthermore, the antenna can be integrated with/onto an outer external case of the mobile internet device **300**.

[0288] The outer external case of the mobile internet device **300** can be fabricated/constructed from a nano-engineered aluminum/magnesium alloy and/or a liquid metal alloy and/or glass.

[0289] The outer external case of the mobile internet device **300** can also be fabricated/constructed from carbon fibers embedded with plastic. Carbon fibers can be inserted into an injection mold of a plastic film and bonded to the molten injection mold of the plastic film, thereby forming a composite material of the carbon fibers and plastic film.

[0290] FIG. **5**B illustrates a cross-section of the display component **1520** of the mobile internet device **300**, which utilizes highly efficient quantum dot light emitting diodes **1520**-A (red, green and blue) incident at an angle with respect to the lower glass substrate **1520**-B. The lower glass substrate **1520**-B has built-in corrugations and waveguides to enable reflection of an incident light from the quantum dot light emitting diodes.

[0291] Table-3 below describes subcomponents required to fabricate/construct the display component **1520**. The critical subcomponents are micro-electromechanical systems micro shutters, which are monolithically integrated with an array of thin-film transistors (TFTs) (e.g., fabricated/constructed on zinc oxide or zinc-indium-tin oxide or graphene oxide).

TABLE-US-00003 TABLE 3 FIG. 5B Legend Description 1520-A Quantum Dot Light Emitting Diodes (Red, Green & Blue) 1520-B Lower Glass Substrate With Built-In Corrugations & Waveguides 1520-C Array of Transparent Lower Pixel Electrodes 1520-D Array of Thin-Film Transistors 1520-E Scanning Transparent Electrodes 1520-F Signal Transparent Electrodes 1520-G Micro-Electromechanical Systems Micro Shutters Monolithically Integrated With A Thin-Film Transistor 1520-H Monolithic Integration of 1520-D, 1520-E, 1520-F & 1520-G 1520-I Transparent Upper Common Electrode 1520-J Upper Glass Substrate

[0292] This can enable an efficient high brightness display component **1520** at lower electrical power consumption, eliminating two (2) polarizer filter films, color filter and liquid crystal. This is substantially compatible with standard display component manufacturing methods/processes.

[0293] Furthermore, a quantum dot white light emitting diode (with a specific thin-film color filter (to transmit only optically filtered red or green or blue light), preferably located below the upper glass substrate) can be used instead of a quantum dot red light emitting diode, a quantum dot green light emitting diode and a quantum dot blue light emitting diode.

[0294] The thin-film transistor **1520**-D located at each pixel can control an image at each pixel of the display component **1520**. However, the thin-film transistor **1520**-D can also have a light sensing circuitry to sense the light reaching the pixel of the display component **1520** from its surroundings—thus enabling a possibility of new user experience with the display component **1520**.

[0295] FIG. **5**C illustrates **1520**-H: a micro-electromechanical systems micro shutter **1520**-G, which can be monolithically integrated with an array of thin-film transistors **1520**-D (e.g., fabricated/constructed on zinc oxide or zinc-indium-tin oxide or graphene oxide).

[0296] The Table-4 below describes subcomponents required to fabricate/construct the micro-electromechanical systems micro shutter **1520**-G, which can be monolithically integrated with an array of thin-film transistors **1520**-D (e.g., fabricated/constructed on zinc oxide or zinc-indium-tin oxide or graphene oxide).

TABLE-US-00004 TABLE 4 FIG. 5C Legend Description 1520-K Thin-Film 1520-L Gate Dielectric 1520-G Micro Shutter 1520-N Source Electrode Or Drain Electrode 1520-M Actuation Electrode 1520-0 Release Layer

[0297] FIG. **5**D illustrates a cross-section of another display component **1520**-T integrated with the solar cell **1240**-A.

[0298] FIG. **5**E illustrates a cross-section of another enabling configuration of the display component **1520** of the mobile internet device **300**. Along with an array of thin-film transistors **1520**-D, the critical element of this configuration is lifted semiconductor quantum-wells layers **1520**-P on a glass substrate (e.g., the glass substrate **1240**-B). Furthermore, the semiconductor quantum-wells layers **1520**-P have both p-metal and n-metal contacts.

[0299] The semiconductor quantum-well layers **1520**-P can be electrically excited by current from a battery. The released energy can be non-radiatively transferred to nanocrystal quantum dots (of various diameters/sizes) **1520**-Q to produce red, green and blue light from an adjacent layer of nanocrystal quantum dots **1520**-Q to enable an efficient color display component **1520**.

[0300] FIG. **5**F illustrates a cross-section of another enabling configuration of the display component **1520** of the mobile internet device **300**. Along with an array of thin-film transistors **1520**-D the critical element of this configuration is epitaxially lifted semiconductor quantum-well layers **1520**-P on a glass substrate (e.g., the glass substrate **1240**-B). Furthermore, the semiconductor quantum-well layers **1520**-P have both p-metal and n-metal contacts.

[0301] The semiconductor quantum-well layers **1520**-P can be electrically excited by current from a battery. The released energy can be non-radiatively transferred to uniformly sized nanocrystal quantum dots **1520**-R to produce white light emission from an adjacent layer of uniformly sized nanocrystal quantum dots **1520**-R. The white light can be filtered by an array of thin-film color filters **1520**-S to enable an efficient color display component **1520**. Details of a quantum dot display have been described/disclosed in U.S. Non-Provisional patent application Ser. No. 12/931,384 entitled “DYNAMIC INTELLIGENT BIDIRECTIONAL OPTICAL ACCESS COMMUNICATION SYSTEM WITH OBJECT/INTELLIGENT APPLIANCE-TO-OBJECT/INTELLIGENT APPLIANCE INTERACTION”, filed on Jan. 31, 2011 (now U.S. Pat. No. 8,548,334, issued on Oct. 1, 2013) (which claims benefit of priority to U.S. Provisional application Ser. No. 61/404,504 entitled “DYNAMIC INTELLIGENT BIDIRECTIONAL OPTICAL ACCESS COMMUNICATION SYSTEM WITH OBJECT/INTELLIGENT APPLIANCE-TO-OBJECT/INTELLIGENT APPLIANCE INTERACTION”, filed on Oct. 5, 2010). Details of various embodiments of the display/holographic display as the display component **1520** of the mobile wearable internet device **300** have been described/disclosed in U.S. Non-Provisional patent application Ser. No. 14/999,601 entitled “SYSTEM AND METHOD OF AMBIENT/PERVASIVE USER/HEALTHCARE EXPERIENCE”, filed on Jun. 1, 2016 (which claims benefit of priority to U.S. Provisional patent application No. 62/230,249 entitled “SYSTEM AND METHOD OF AMBIENT/PERVASIVE USER/HEALTHCARE EXPERIENCE”, filed on Jun. 1, 2015).

[0302] FIG. **6** illustrates a block diagram of the sketch pad (with write/erase options) electronic module **1560** with the stylus **1580**. The sketch pad electronic module **1560** is a multilayer device, having a transparent (e.g., indium tin oxide or graphene) input matrix **1680**, below the transparent input matrix **1680**, there is the display component (e.g., a liquid crystal or graphene based display component) **1520** and below which there is an electronic (scan, drive and display memory) circuitry **1700**. **1560**-A is an integrated electronic device, excluding the stylus **1580**.

[0303] The stylus **1580** can be formed in the shape of a pencil from silicon rubber impregnated with metal particles.

[0304] As the stylus **1580** writes over the transparent input matrix **1680**, it can capacitively couple with the transparent input matrix **1680**. Thus, if there is a change in the capacitance, the change in the capacitance can be sensed by the electronic circuitry **1700**. The electronic circuitry **1700** can be electrically coupled with a switch **1720**. Utilizing the switch **1720**, the sketch pad electronic module **1560** can be operated in both write and erase modes.

[0305] FIG. **7** illustrates a block diagram of the sketch pad electronic module **1560**, where the electrical coupling between the transparent input matrix **1680**, display component **1520**, electronic circuitry **1700** and switch **1720** are described. Furthermore, a hand-writing recognition algorithm and/or a pattern recognition algorithm **1740** can enhance the performance of the sketch pad electronic module **1560**.

[0306] FIG. **8**A illustrates a block diagram of the personal awareness assistant miniature electronic module **1620**, which integrates: the storage/memory **520** (however, the storage/memory **520** can also be replaced by the general storage electronic module **520**-C), the sensor **1120**, the medium performance microprocessor **1260**, the camera **1480**, the voice-to-text-to-voice conversion algorithm **1640**, two (2) microphones **1760**, a scrolling audio recording buffer **1780** and a voice recognition algorithm **1800**.

[0307] The personal awareness assistant miniature electronic module **1620** can always be on. It can passively listen to what the user **160** says and can respond to a particular context and situation. For example: the user **160** can hear about a product and the user **160** can create a reminder by speaking to the personal awareness assistant miniature electronic module **1620**. The user **160** can transmit that information from the personal awareness assistant miniature electronic module **1620** to the social wallet **100** via the electronic social wallet electronic module **280** and/or mobile internet device **300** for further processing and/or fulfillment. After processing the information from the personal awareness assistant miniature electronic module **1620**, the social wallet **100** can then deliver a real time location based coupon(s) to the mobile internet device **300**, by measuring the user's **160** location information by utilizing an indoor/outdoor location measurement miniature electronic module **1440** of the mobile internet device **300**.

[0308] Optionally the personal awareness assistant miniature electronic module **1620** can be a standalone miniature electronic module (but it can be plugged in or integrated with the social wallet electronic module **280**/mobile internet device **300**).

[0309] For example, when the user **160** is introduced to someone, the personal awareness assistant miniature electronic module **1620** can automatically recognize and may take a low-resolution photo. Once, the mobile internet device **300** collects information, it can automatically categorize the information into a pre-designated database with audio, digital image, time/date stamp and global positioning system location. Because the data is stored contextually, the information retrieval can be straightforward. In response to a simple voice command inquiry, such as “whom did I meet on Apr. 15, 2009 at 12 PM”? the personal awareness assistant miniature electronic module **1620** can bring up the appropriate information about that specific person. Thus, the mobile internet appliance is context-aware.

[0310] Furthermore, the voice recognition algorithm **1800** can enhance the capability of the personal awareness assistant miniature electronic module **1620**.

[0311] Additionally, a face recognition algorithm can enhance the capability of the personal awareness assistant miniature electronic module **1620**.

[0312] FIG. **8**B illustrates an application of the personal awareness assistant miniature electronic module **1620** of the mobile internet device **300**. In step **4062**, the user finds a product (e.g., on the radio news). In step **4063**, the personal awareness assistant miniature electronic module **1620** records that product information with the user's consent. In step **4064**, the personal awareness assistant miniature electronic module **1620** transmits the product information to the social wallet **100**. In step **4065**, the social wallet **100** acknowledges the received (product) information from the personal awareness assistant miniature electronic module **1620**. In step **4066**, the social wallet **100** processes the received (product) information from the personal awareness assistant miniature electronic module **1620**. In step **4067**, the social wallet **100** determines the location of the mobile internet device **300** in real time. In step **4068**, the social wallet **100** determines the local stores, where the product can be found. In step **4069**, the social wallet **100** directs (turn by turn) the mobile internet device **300** in real time regarding the location of a specific or closest store, where the product can be found. In step **4070**, the user **160** pays for the product using the mobile internet device **300** (the mobile internet device **300** can be integrated with the social wallet electronic module **280**). Furthermore, the social wallet electronic module **280** is integrated with the near-field communication miniature electronic module **620**.

[0313] As the social wallet **100** can learn or relearn the user's preferences, the unified algorithm **320** can render intelligence based on the user's preferences, by utilizing the intelligence rendering algorithm **400** and the self-learning (including relearning) algorithm **420**.

[0314] Similarly, the mobile internet device **300** can also learn or relearn the user's preferences, by utilizing the algorithm **1660**. The algorithm **1660** includes: (a) a physical search algorithm, (b) an algorithm-as-a-service, (c) an intelligent rendering algorithm (e.g., artificial intelligence, behavior modeling, data interpretation, data mining, fuzzy logic, machine vision, natural language processing, neural network, pattern recognition and reasoning modeling) and (d) a self-learning (including relearning) algorithm. It should be noted that the self-learning (including relearning) algorithm can include a self-learning artificial intelligence algorithm and/or a self-learning neural network algorithm.

[0315] The personal awareness assistant miniature electronic module **1620** is context-aware. Thus, the mobile internet device **300** is also context-aware.

[0316] FIG. **9** illustrates a block diagram of a first system-on-chip **1820**-A, integrating a microprocessor **1360**, a graphic processor unit (GPU) **1360**-A and an internet security algorithm **1400**.

[0317] FIG. **10** illustrates a block diagram configuration of a nano-transistor **1840**-A. A lower nanowire array of a switchable active material (e.g., silicon or germanium) **1840**-B sandwiched between contacts (e.g., nickel silicide for silicon or nickel germanide for germanium) **1840**-C can be fabricated/constructed. An upper array of gate metal **1840**-D enclosed within an insulating shell **1840**-E can be fabricated/constructed. Voltage applied via the gate metal **1840**-D can switch the active material **1840**-B from an off state to an on state.

[0318] FIG. **11** illustrates a block diagram configuration of a second system-on-chip **1820**-B, which integrates a microprocessor **1360**-B, (where the microprocessor **1360**-B is based on nano-transistors **1840**-A), the graphic processor unit **1360**-A and the internet security algorithm **1400**.

[0319] Molybdenite (MoS.sub.2) is a two-dimensional crystal with a natural bandgap. It is suitable for production of digital integrated circuits. A reduction in bandgap and/or increase in mobility of molybdenite can be achieved by an addition of lithium (Li) ions.

[0320] Graphene is also a two-dimensional crystal with a higher carrier mobility, as well as lower noise. It has the ideal properties to be an excellent component of integrated circuits. Graphene epitaxially grown on silicon carbide (SiC) may be suitable for production of integrated circuits.

[0321] A graphene variant called graphene has hydrogen atoms attached to the carbon lattice in insulating layers.

[0322] Graphyne is a one-atom-thick sheet of carbon that resembles graphene, except in its two-dimensional framework of atomic bonds, which contains triple bonds in addition to double bonds. Graphyne has a graphene-like electronic structure resulting in effectively massless electrons due to Dirac Cones. All electrons are travelling at roughly the same speed (about 0.3% of the speed of light). This uniformity leads to conductivity greater than copper. Graphyne has a capability of self-modulating its electronic properties, which means that it could be used as a semiconductor practically as-is, without requiring any non-carbon dopant atoms to be added as a source of electrons, as non-carbon dopants may be required for graphene. Furthermore, graphyne crystal

structure allows electrons to flow in just one direction.
[0323] A first lower parallel array of nanoscaled metal (platinum) wires can be fabricated/constructed onto a substrate. A titanium oxide-titanium dioxide thin film can be deposited on the first lower parallel array of nanoscaled metal wires. A second upper parallel array of nanoscaled metal (platinum) wires can be fabricated/constructed on top of the titanium oxide-titanium dioxide thin film. The second upper parallel array is typically fabricated/constructed perpendicular to the first lower parallel array.
[0324] A memristor of a titanium oxide-titanium dioxide oxide junction, can be formed when the first lower parallel array of nanoscaled metal (platinum) wires cross the second upper parallel array of nanoscaled metal (platinum) wires. A memristor is less than 50 microns×50 microns in size. A memristor is a two-terminal nanoscaled non-linear passive switching element, whose resistance changes depending on the amount, direction and duration of voltage applied on it. But whatever its past state or resistance was, it freezes at that state, until another voltage is applied to change it. It has a variable resistance and can retain the resistance even when the electrical power is switched off. It is similar to a transistor, used to store data in a flash memory. Since a memristor is a two-terminal microscaled/nanoscaled passive switching element, it can be built on top of transistors to electrically power it up.
[0325] Phase-change memory (e.g., germanium-antimony-tellurium) has been used in optical information technologies (e.g., a DVD) and non-volatile memory applications. Furthermore, a phase-change memory material based switching element can be used instead of a memristor. The phase-change memory material based switching element exploits a unique switching behavior of a phase-change material between amorphous (high resistivity) and crystalline (low resistivity) material states with the application of electrical pulses by titanium nitride top electrode and titanium nitride-tungsten bottom electrode to generate the required joule heating for the phase transformation.
[0326] Furthermore, a dense local network of switching elements **1840***s* (e.g., based on memristors and/or phase-change memory material based switching elements) can be monolithically integrated with transistors fabricated/constructed on a semiconductor (e.g., silicon or germanium or silicon-germanium) and/or nano-transistors fabricated/constructed on a semiconductor (e.g., silicon or germanium or silicon-germanium) and/or transistors fabricated/constructed on two-dimensional crystal.
[0327] FIG. **12** illustrates the switching element **1840** based on the memristor and/or phase-change memory material.
[0328] Thus, transistors (fabricated/constructed on a semiconductor and/or two-dimensional crystal) with integrated switching elements **1840** can be utilized to fabricate/construct a reconfigurable (and with lower electrical power consumption) advanced microprocessor **1360**-C.
[0329] FIG. **13** illustrates a block diagram configuration of a third system-on-chip **1820**-C, which integrates an advanced microprocessor **1360**-C, the graphics processor unit **1360**-A and the internet security algorithm **1400**.
[0330] In the human brain, neurons are connected to each other through programmable junctions called synapses. The synaptic weight modulates how signals are transmitted between neurons and can in turn be precisely adjusted by an ionic flow through the synapse.
[0331] The switching element **1840** is a non-linear resistive device with an inherent memory and similar to a synapse. It is a two-terminal device whose conductance can be modulated by an external stimulus with the ability to store (memorize) the new information. The switching element **1840** can bring data close to computation without a lot of electrical power consumption, as a biological neural system does.
[0332] FIG. **14** illustrates an embodiment of a neural network microprocessor **1360**-D. **1360**-D integrates the switching element **1840** based synapses, complementary metal-oxide semiconductor (CMOS) pre-neurons **1860** (fabricated/constructed on a semiconductor and/or two-dimensional

crystal) and complementary metal-oxide semiconductor post-neurons **1880** (fabricated/constructed on a semiconductor and/or two-dimensional crystal).

[0333] FIG. **15** illustrates a block diagram configuration of a fourth system-on-chip **1820**-D, which integrates an advanced microprocessor (based on a neural network) **1360**-D, the graphic processor unit **1360**-A and the internet security algorithm **1400**.

[0334] FIG. **16** illustrates another embodiment of an advanced microprocessor **1360**-E (based on a neural network and nano-transistors **1840**-A). **1360**-E integrates the switching element **1840** based synapses, complementary metal-oxide semiconductor pre-neurons **1860** (fabricated/constructed on a semiconductor and/or two-dimensional crystal), complementary metal-oxide semiconductor post-neurons **1880** (fabricated/constructed on a semiconductor and/or two-dimensional crystal) and nano-transistors **1840**-A.

[0335] Alternatively, another embodiment of an advanced microprocessor **1360**-E (based on a neural network and nano-transistors **1840**-A). **1360**-E integrates the switching element **1840** based synapses, complementary metal-oxide semiconductor pre-neurons **1860** (fabricated/constructed on an electrically induced (e.g., voltage or current) phase change material (e.g., germanium-antimony-tellurium (GST) or $Ag_4In_3Sb_{67}Te_{26}$ (AIST))), complementary metal-oxide semiconductor post-neurons **1880** (fabricated/constructed on a semiconductor and/or two-dimensional crystal) and nano-transistors **1840**-A.

[0336] Alternatively, another embodiment of an advanced microprocessor **1360**-E (based on a neural network and nano-transistors **1840**-A). **1360**-E integrates the switching element **1840** based synapses, complementary metal-oxide semiconductor pre-neurons **1860** (fabricated/constructed on an electrically induced (e.g., voltage or current) phase transition material (e.g., vanadium dioxide (VO_2)), complementary metal-oxide semiconductor post-neurons **1880** (fabricated/constructed on a semiconductor and/or two-dimensional crystal) and nano-transistors **1840**-A.

[0337] Alternatively, another embodiment of an advanced microprocessor **1360**-E (based on a neural network and nano-transistors **1840**-A). **1360**-E integrates the switching element **1840** based synapses, complementary metal-oxide semiconductor pre-neurons **1860** (fabricated/constructed on an optically induced phase transition material (e.g., vanadium dioxide (VO_2)), complementary metal-oxide semiconductor post-neurons **1880** (fabricated/constructed on a semiconductor and/or two-dimensional crystal) and nano-transistors **1840**-A. It should be noted that optically induced phase transition material may enable a faster switching memristor (hence, a photonic neural learning processor).

[0338] Details of a photonic neural learning processor have been described/disclosed in U.S. Non-Provisional patent application Ser. No. 16/602,404 entitled "SYSTEM AND METHOD OF AMBIENT/PERVASIVE USER/HEALTHCARE EXPERIENCE", filed on Sep. 28, 2019 and its related priority patent applications.

[0339] Generally, a phase transition material is a solid material, wherein its lattice structure can change from a particular form to another form, still remaining crystal-graphically solid.

[0340] Generally, a phase change material is a material, wherein its phase can change from a solid to liquid, or its phase can change from an amorphous to crystalline, or crystalline to amorphous.

[0341] FIG. **17** illustrates a block diagram configuration of a fifth system-on-chip **1820**-E, which integrates an advanced microprocessor (based on a neural network and nano-transistors **1840**-A) **1360**-E, the graphic processor unit **1360**-A and the internet security algorithm **1400**.

[0342] A quantum dot memory (e.g., an array of silicon/silicon-germanium quantum dots embedded in epitaxial rare-earth oxide gadolinium oxide (Gd_2O_3) grown on a silicon substrate of (111) crystal orientation) can be fabricated/constructed with the system-on-chip **1820**-A/B/C/D/E.

[0343] Additionally, a cross-point memory can be fabricated/constructed in multiple layers to form three-dimensionally. The three-dimensional cross-point memory can be fabricated/constructed directly on top of the system-on-chip **1820**-A/B/C/D/E. Furthermore, the optical interconnect (optical layers) can also be integrated with the system-on-chip **1820**-A/B/C/D/E. Details on memristors and an optical interconnect have been described/disclosed in U.S. Non-Provisional patent application Ser. No. 14/999,601 “SYSTEM entitled AND METHOD OF AMBIENT/PERVASIVE USER/HEALTHCARE EXPERIENCE”, filed on Jun. 1, 2016 (which claims benefit of priority to U.S. Provisional patent application No. 62/230,249 entitled “SYSTEM AND METHOD OF AMBIENT/PERVASIVE USER/HEALTHCARE EXPERIENCE”, filed on Jun. 1, 2015).

[0344] The system-on-chips **1820**-A/B/C/D/Es, optically interconnected can enable the learning (relearning) computer to store and process massive datasets. Furthermore, the system-on-chips **1820**-A/B/C/D/Es, optically interconnected and a neural network based algorithm(s) can enable supervised, unsupervised and semi-supervised learning. The advanced microprocessors **1360**-D and **1360**-E can have Cog Ex machines/Machine OS, as an operating algorithm/system.

[0345] FIGS. **18**A, **18**B and **18**C illustrate a block diagram of process flow for integrating two (2) two-dimensional crystals on an insulator on a semiconductor substrate (e.g., silicon or germanium or silicon-germanium). Graphene/graphene/graphyne is grown on a substrate X (e.g., graphene can be epitaxially grown on silicon carbide substrate).

[0346] Graphene/graphene/graphyne can be patterned with a photoresist and reactive ion beam (RIE) etching. Graphene/graphene/graphyne can by be bonded and detached poly(dimethylsiloxane) (PDMS) onto an insulator on a semiconductor substrate. Thus, the above semiconductor fabrication process/method enables integration of one or two two-dimensional crystals on an insulator on a semiconductor substrate for further circuit fabrication.

[0347] For efficient thermal management of the system-on-chip **1820**-A/B/C/D/E for the mobile internet device **300**, thermal resistance must be minimized at all material interfaces and materials with closely matching thermal expansion coefficients must be used.

[0348] FIG. **19** illustrates that the circuit side of the system-on-chip **1820**-A/B/C/D/E can be flip-attached or flip-bonded on an array of thermoelectric films (both n-type and p-type) **1920**_s_ with a built-in nano-structured surface **1900** for active cooling.

[0349] About ten times (10×) heat transfer can be realized by creating a nano-structured surface (e.g., zinc oxide/graphene nano-structured surface) **1900** on the thermoelectric film **1920**. However, significant thermoelectric efficiency can be gained by fabricating a quantum wire/quantum dot based thermoelectric film **1920**, transitioning from a two-dimensional superlattice.

[0350] Furthermore, the thermoelectric film can be attached or bonded on a thermal pillar (e.g., copper material) **1940**. The thermal pillar **1940** is about 250 microns in diameter and 50 microns in height. The thermal pillar (e.g., copper material) **1940** can be attached or bonded on a thermal via **1960** on a printed circuit board **1980** with a cooling module **2000**.

[0351] FIG. **20**A illustrates the cooling module **2000**, which can be attached or bonded with the printed circuit board **1980** to disperse the heat from the system-on-chip **1820**-A/B/C/D/E. The cooling module **2000** consists of an array of mini cooling modules **2000**-As.

[0352] FIG. **20**B illustrates the mini cooling module **2000**-A. The mini cooling module **2000**-A has an array of negative voltage biased tips (e.g., tips fabricated/constructed from boron nanotube/carbon nanotube/amorphous diamond/tungsten) **2020**_s_, which is placed just below a hole (e.g., about 100 microns in diameter) **2040** of positive voltage biased surface (e.g., tungsten/two-dimensional crystal material (e.g., graphene)) **2060**. Electrons emitted from the negative voltage biased array of tips **2020**_s_ can escape through the hole **2040** and ionize the gas molecules within the boundaries of a heat sink (e.g., the heat sink can be fabricated/constructed from materials such as aluminum/silicon/copper/carbon nanotube-copper composite/two-dimensional crystal material (e.g., graphene)/diamond) **2080**. By switching the voltage polarity of the heat sink **2080**, a moving ionized gas cloud can disperse the heat from the printed circuit board **1980**.

[0353] However, it is desirable that an array of tips **2020**_s_ emits electrons at a much lower voltage (e.g., 10 volts).

[0354] FIG. **20**C illustrates an array of nano-sized tungsten tips **2020***s*, which can be fabricated/constructed on tungsten substrate **2040**-A. The tungsten tips **2020***s* can be surrounded by an insulator **2040**-B. The nano-sized tungsten tip **2020** can be decorated with a monolayer of material **2020**-A (e.g., diamond deposited by low temperature electron cyclotron resonance-chemical vapor deposition (ECR-CVD) or gold deposited by RF magnetron sputtering) to enable electrons to emit at a much lower voltage (e.g., at 10 volts) through the hole **2040**, where the hole **2040** can be fabricated/constructed from a tungsten material.
[0355] To achieve faster connectivity between the system-on-chip **1820**-A/B/C/D/E, an optical interconnection is preferable to an electrical interconnection.
[0356] FIG. **21**A illustrates a block diagram of an interconnection between the system-on-chip **1820**-A/B/C/D/E (via optics) on the printed circuit board **1980**.
[0357] The Table-5 below describes subcomponents required to fabricate/construct the interconnection between the system-on-chip **1820**-A/B/C/D/E (via optics) on the printed circuit board **1980**.
TABLE-US-00005 TABLE 5 FIG. 21A Legend Description 1820-A/B/C/D/E System-On-Chip 2100 Complementary Metal-Oxide Semiconductor Serializer 2120 Complementary Metal-Oxide Semiconductor Driver 2140 Directly Modulated Vertical Cavity Laser 2140-A Vertical Cavity Laser Integrated With Electro-Optical Modulator 2160 Two-Dimensional Photonic Crystal Wavelength Division Multiplexer On Silicon-On-Insulator (SOI) 2180 Silicon-On-Insulator Waveguide 2200 Reconfigurable Optical Switch On Silicon-On-Insulator 2220 Silicon-On-Insulator Two-Dimensional Photonic Crystal Wavelength Division Demultiplexer 2240 Photodetector 2260 Complementary Metal-Oxide Semiconductor Amplifier 2280 Complementary Metal-Oxide Semiconductor Deserializer 1820-A/B/C/D/E System-On-Chip
[0358] Electrical outputs from the system-on-chip (e.g., **1820**-A/B/C/D/E) are serialized by a complementary metal-oxide semiconductor serializer **2100**. The outputs of a complementary metal-oxide semiconductor serializer **2100** can be utilized as inputs to an array of complementary metal-oxide semiconductor drivers **2120***s*. Correspondingly, the array of complementary metal-oxide semiconductor drivers **2120***s* can activate an array of directly modulated (in intensity) vertical cavity surface emitting lasers **2140***s* or an array of vertical cavity surface emitting lasers, which are monolithically integrated with electro-optic modulators **2140**-As.
[0359] The modulated wavelengths of the directly modulated vertical cavity surface emitting lasers **2140***s* or vertical cavity surface emitting lasers with monolithically integrated with electro-optic modulator **2140**-As can be combined on wavelengths (or colors) by a silicon-on-insulator two-dimensional photonic crystal wavelength division multiplexer **2160**.
[0360] The wavelengths can be propagated by a silicon-on-insulator waveguide **2180** and if necessary, can be reconfigured by a reconfigurable optical switch **2200** on silicon-on-insulator.
[0361] The outputs of the silicon-on-insulator waveguide **2180** or reconfigurable optical switch **2200** on silicon-on-insulator can be decombined on wavelengths by a silicon-on-insulator two-dimensional photonic crystal wavelength division demultiplexer **2220**.
[0362] Furthermore, the wavelengths outputs (of the silicon-on-insulator two-dimensional photonic crystal wavelength division demultiplexer **2220**) can be received by an array of photodetectors (e.g., P-i-N photodetectors) **2240***s*, an array of complementary metal-oxide semiconductor amplifiers **2260***s*, then as electrical inputs to a complementary metal-oxide semiconductor deserializer **2280** and finally as electrical inputs to another system-on-chip (e.g., **1820**-A/B/C/D/E).
[0363] FIG. **21**B illustrates details of the silicon-on-insulator waveguide **2180**, silicon-on-insulator vertical coupler gratings **2300** and directly modulated vertical cavity surface emitting laser **2140** or the vertical cavity surface emitting laser with monolithically integrated electro-optic modulator **2140**-A.
[0364] FIG. **21**C illustrates details of the silicon-on-insulator waveguide **2180**, silicon-on-insulator vertical coupler gratings **2300** and photodiode **2240**. The shape of the silicon-on-insulator

waveguide **2180** (fabricated/constructed on an oxide buffer layer **2320** on the silicon substrate **2340**) can be adiabatically tapered at proximity of the silicon-on-insulator vertical coupler gratings **2300**. The silicon-on-insulator vertical coupler gratings **2300** can be shaped linear or curved.
[0365] Furthermore, the silicon-on-insulator two-dimensional photonic crystal wavelength division multiplexer **2160**, silicon-on-insulator waveguide **2180**, reconfigurable optical switch **2200** on silicon-on-insulator and silicon-on-insulator two-dimensional photonic crystal wavelength division demultiplexer **2220** can be embedded within an etched area of polymer layers of the printed circuit board **1980**. An optical mode match between the silicon-on-insulator waveguide **2180** and a polymer waveguide (by utilizing a polymer layer of the printed circuit board **1980**) can be fabricated/constructed. Also, the etched area can be buried within the printed circuit board **1980**. Alternatively, a polymer (e.g., polyimide) waveguide of the printed circuit board **1980** can be utilized instead of the silicon-on-insulator waveguide **2180**.
[0366] FIG. **21**D illustrates a cross section of **2140**-A: a vertical cavity surface emitting laser, which can be monolithically integrated with an electro-optic modulator. Similarly, a micro-electromechanical systems tunable vertical cavity surface emitting laser (preferably a quantum dot vertical cavity surface emitting laser) can also be monolithically integrated with an electro-optic modulator.
[0367] FIG. **22**A illustrates an embodiment of a (optically enabled) Super System on Chip. Specifically, (an optically enabled) Super System on Chip's input/output can be coupled by a Mach-Zehnder interferometer, wherein the Mach-Zehnder interferometer can include a phase transition material or a phase change material or a polymeric material (wherein a phase transition material is electrically and/or optically controlled, but a phase change material is electrically or optically controlled), wherein the Mach-Zehnder interferometer can be coupled by a low-loss first optical (routing) waveguide in either two-dimensions/three-dimensions, wherein the low-loss first optical waveguide can be coupled by a semiconductor optical amplifier (for nonlinear optical processing) in either two-dimensions/three-dimensions. It should be noted that a three-dimensional arrangement can include a vertical coupling.
[0368] It should be noted that memristors can be replaced by super memristors. Each super memristor can include (i) a resistor, (ii) a capacitor and (iii) a memristor (e.g., a phase transition/phase change material based memristor).
[0369] A phase transition material based memristor can be electrically and/or optically controlled. But a phase change material based memristor can be electrically or optically controlled.
[0370] A super memristor can generally mimic a set of neural activities (such as simple spikes, bursts of spikes and self-sustained oscillations with a DC voltage as an input signal)—which can be used for a neuromorphic/neural processing/computing architecture. Furthermore, each super memristor can be electrically/optically controlled.
[0371] It should be noted that the semiconductor optical amplifier can be replaced by a second optical waveguide containing a nonlinear optical material (e.g., chalcogenide As.sub.2S.sub.3) for nonlinear optical processing. This is generally illustrated in FIG. **22**B.
[0372] Furthermore, the semiconductor optical amplifier can also include a second optical waveguide containing a nonlinear optical material for advanced nonlinear optical processing. This is generally illustrated in FIG. **22**C.
[0373] FIG. **22**D illustrates a two-dimensional representation of a (optically enabled) Super System on Chip, wherein an input and/or an output of the (optically enabled) Super System on Chip can be coupled with a Mach-Zehnder interferometer, wherein then the Mach-Zehnder interferometer can be coupled with a first optical waveguide (FW) in a two-dimensional arrangement, wherein the first optical waveguide can be coupled optically with a semiconductor optical amplifier. Furthermore, the first optical waveguide can be coupled optically with a second optical waveguide in the two-dimensional arrangement, wherein the second optical waveguide may include a nonlinear optical material. The (optically enabled) Super System on Chip can be coupled optically with a laser (LD)

and one or more photodiodes (PDs), denoted as PD**1** and PD**2** in FIG. **22**D. It should be noted there are both (i) optical connections via optical waveguides and (ii) electrical connections in FIG. **22**D.

[0374] It should be noted that a semiconductor optical amplifier can be replaced by an optical resonator.

[0375] It should be noted that the above two-dimensional representation of the (optically enabled) Super System on Chip (in FIG. **22**D) can be also realized in a three-dimensional arrangement (wherein the three-dimensional arrangement can include a vertical coupling via precision chip/wafer level bonding or via an optical equivalent of an electrical via hole).

[0376] In general, the Super System on Chip can be communicatively interfaced with (i) a first set of computer implementable instructions to process an audio (signal) input (e.g., voice), (ii) a second set of computer implementable instructions to analyze and/or interpret contextual data, depending on the context of data and (iii) a third set of computer implementable instructions in artificial neural networks, wherein the artificial neural networks may further include either a transformer model or a diffusion model (which may be further augmented with an evolutionary based algorithm and/or a game theory based algorithm and/or Poisson flow generative model++ based algorithm).

[0377] It should be noted that the second set of computer implementable instructions and the third set of computer implementable instructions may enable (i) self-learning and/or (ii) a (personal) artificial intelligence based self-learning assistant, (e.g., FIG. **23**).

[0378] Furthermore, the first set of computer implementable instructions, the second set of computer implementable instructions and the third set of computer implementable instructions can be stored either (i) locally with the Super System on Chip (including optically enabled Super System on Chip) and/or the system-on-chip or (ii) in a remote/cloud server (which can be accessed by the Super System on Chip and/or the system-on-chip over the internet from the remote/cloud server).

[0379] FIG. **23** illustrates an embodiment of a (personal) artificial intelligence based self-learning assistant in a block diagram, wherein a sensor/biosensor/biological lab-on-a-chip/bioobject's data input, audio (signal) input in natural language and contextual data are fed into an artificial neural network to analyze and/or interpret (i) sensor data, (ii) audio (signal) input in natural language and (iii) contextual data utilizing the Super System on Chip (including optically enabled Super System on Chip) and/or the system-on-chip. It should be noted that the system-on-chip can include one or more (i) graphic processors and (ii) on-sensor processing circuits. An on-sensor processing circuit can include one or more digital signal processors (DSPs). Furthermore, the system-on-chip can include one or more multipliers of matrices.

[0380] The artificial neural network may further include a transformer model or a diffusion model (which may be further augmented with an evolutionary based algorithm and/or a game theory based algorithm and/or Poisson flow generative model++ based algorithm).

[0381] For example, an autonomous artificial intelligence agent can plan a vacation with high accuracy rate on each step of a multistep process or very good error correction to get anything valuable out of an autonomous artificial intelligence agent that has to take lots of steps. This is contrary to standard artificial intelligence based processes that are run only when a user triggers them and only to accomplish a specific result and then standard artificial intelligence based processes stop. Characteristics of an autonomous artificial intelligence agent are (i) autonomy, (2) continuous learning and (3) reactive and proactive in an environment.

[0382] Furthermore, (i) the artificial neural networks (including Kolmogorov-Arnold based neural network architecture and evolutionary algorithm based instructions), (ii) natural language processing and (iii) a collection of declarative knowledge based computer implementable instructions (to enable common sense) can be used to design an autonomous artificial intelligence agent.

[0383] An autonomous artificial intelligence agent is a set of computer implementable instructions (an algorithm) and it can have multimodal inputs via text, video and voice.

[0384] An autonomous artificial intelligence can analyze the collected data (e.g., from sensors/biosensors) to make informed decisions and take an action or a series of actions to achieve its goals.

[0385] An autonomous artificial intelligence can be stored in a non-transitory storage media, located either locally on the intelligent subsystem or in the cloud server.

[0386] Thus, the (personal) artificial intelligence based self-learning assistant may be able to recommend useful information proactively without asking/searching for such information explicitly (via a Recommendation Engine).

[0387] Following FIG. **23**, the (personal) artificial intelligence based self-learning assistant can include one or more computer implementable instructions including in artificial neural networks (which can include a transformer model or a diffusion model and may also be augmented with evolutionary based instructions and/or game theory based instructions and/or Poisson flow generative model++ based instructions.

[0388] Thus, one or more computer implementable instructions can be stored either (i) locally with the Super System on Chip (including optically enabled Super System on Chip) and/or the system-on-chip or (ii) in a remote/cloud server (which can be accessed by the Super System on Chip and/or the system-on-chip over the internet from the remote/cloud server) for the following: [0389] Anticipate a user's needs based on context (e.g., based on an audio signal in natural language or text), [0390] Help with daily tasks/reminders, [0391] Schedule meetings, [0392] Edit photos/images based on context, [0393] Identify photos/images, [0394] Help with nighttime imaging (photography), [0395] Help with smart selfie blurs, [0396] Help with smart selfie blurs with an augmented reality image, [0397] Create custom emojis based on context, [0398] Translate voice memos into another language, [0399] Translate voice memos into text/email into another language, [0400] Generate/phone call on behalf of a user, [0401] Generate/send voicemails on behalf of a user, [0402] Generate/send emails on behalf of a user, [0403] Generate/send contacts on behalf of a user, [0404] Generate/summarize daily voice memos, [0405] Generate/summarize daily action notes, [0406] Remind of daily meetings, based email/voicemail context, [0407] Suggest cooking recipe based on the food images, [0408] Suggest future action items based on context, [0409] Share live events (e.g., Lebron James blocking the shot of the Golden State Warriors' Andre Iguodala), [0410] Annotate/comment on live events (e.g., Lebron James blocking the shot of the Golden State Warriors' Andre Iguodala—with annotation “Witnessing Greatness”), [0411] Share annotated/commented live events, [0412] Compose custom text, images and videos based on context, [0413] Compose custom songs based on context, [0414] Compose similar songs by listening/understanding/audio processing a particular song, [0415] Generate immersive concerts based on context, [0416] Generate immersive games based on context, [0417] Generate/summarize information from the internet search based on a user preference/interest (including an implied preference/interest) without having to open and read multiple links on the internet, [0418] Generate/enable a tipping point alert of any expected virus propagation or any expected natural disaster (e.g., an earthquake or a fire), [0419] Generate immersive digital companion, [0420] Generate a digital twin, [0421] Constantly learn from data and interaction with a user and sensors-including implicit feedback from a user, Infer user interests-including implicit interests and enable proactive assistance utilizing contextual reasoning. For example, detect health related problems/issues of a user based on inputs from biosensors (placed on or implanted on a user) and call 911 emergency automatically with the GPS coordinates of a user, without any intervention of a user, [0422] Besides the internet, the (personal) artificial intelligence based self-learning assistant can search/scan other resources such as various search engines (e.g., Bing, Google, Yahoo and Yelp), expert databases, data from existing Question & Answer forums (e.g., ChaCha) and answers drawn from real-time applications that will ask relevant people if they know answer. What makes the Question & Answer forums powerful is that they keep track of each and every question and answer pairing ever asked and every answer ever given. For example, as noted before if an elderly user living alone in a home, suddenly experiences a heart attack, a biosensor/biological lab-on-a-chip on the elderly user can detect such health problem and call 911 emergency automatically with the location coordinates of the elderly user, without any intervention of the elderly user. [0423] The (personal) artificial intelligence (which can be sensor-aware and/or context-aware) based self-learning assistant (an interface for computer implementable instructions) may be considered as a digital personal assistant (DPA), (which can be sensor-aware and/or context-aware) and then coupled/communicatively interfaced with a chatbot interface (that generally mimics human level natural conversation) and one or more computer implementable instructions (stored in one or more non-transitory storage media, located either locally on the intelligent subsystem or in a cloud server) to provide a recommendation (including an implied/inferred/interpreted recommendation) based on a user's interest/preference that was received at the intelligent subsystem. A chatbot interface with natural language processing and semantic analyzer may allow having a natural conversation, as opposed to using an inherent keyboard for everything. Semantic analysis is a subfield of natural language processing that attempts to understand the meaning of natural language of the given text, while considering the context, logical structuring of sentences and grammar. Semantic analyzer can understand the context of natural language and/or detect emotion/sarcasm and extract valuable information from unstructured data, achieving human-level accuracy. Utilizing library learning (e.g., the library induction from language observations (LILO) and/or the action domain acquisition (ADA)) the chatbot can learn, perceive, reason and represent human level natural conversation. The library induction from language observations—a neurosymbolic framework that can synthesize, compress and document computer codes and it can utilize a large language model (LLM) to cut and combine codes into libraries of readable and reusable algorithms. Similarly, the action domain acquisition (natural language that guides artificial intelligence based algorithm for multi-step task planning) can enable sequential decision making with artificial intelligence based algorithms. [0424] Such a digital assistant that not only responds to a user's (input) instructions but also is able to learn, adapt and take (meaningful) autonomous actions to solve one or more problems on the user's behalf. [0425] Furthermore, the (personal) artificial intelligence based self-learning assistant can be communicatively interfaced with one or more computer implementable instructions (stored in one or more non-transitory storage media, located either locally on the intelligent subsystem or in a cloud server) to learn through an audio signal (e.g., voice) (via computer implementable instructions in natural language understanding/processing—stored in one or more non-transitory storage media, located either locally on the intelligent subsystem or in a cloud server), text, video, sensor data (e.g., from a sensor/biosensor) and (gained) experience by solving iterative problems. [0426] Furthermore, the (personal) artificial intelligence based self-learning can enhance utility of the (personal) artificial intelligence based self-learning assistant in audio signal enabled intelligent (self-learning) computing and enable the following: Sentiment analysis, Learning/adapting to user preference and intent (including an implied/inferred/interpreted intent), Understanding user behavior pattern and responding proactively. [0427] The (personal) artificial intelligence based self-learning assistant can be communicatively interfaced with an autonomous artificial intelligence agent, (e.g., it can plan a vacation) with high accuracy rate on each step of a multistep process or very good error correction to get anything valuable out of an autonomous artificial intelligence agent that has to take lots of steps. This is contrary to standard artificial intelligence based processes that are run only when a human triggers them and only to accomplish a specific result and then standard artificial intelligence based processes stop. Characteristics of an autonomous artificial intelligence agent are (i) autonomy, (2) continuous learning and (3) reactive and proactive in an environment. [0428] The (personal) artificial intelligence based self-learning assistant communicatively interfaced with an autonomous artificial intelligence agent can book on the next flight for a user, when the (personal) artificial intelligence based self-learning assistant finds out from the internet and other resources that the previous flight is canceled. The (personal) artificial intelligence based self-learning assistant can then communicate with a user's family about the delay in arrival, or newly booked flight and then notify/reorder the airport shuttle/taxicab accordingly to pick up a user from the airport. [0429] The (personal) artificial intelligence based self-learning assistant interfaced with an autonomous artificial intelligence agent can enable an intelligent and intuitive internet search interface (Product/Service Matching/Searching a Buyer/User based on a User Profile, as opposed to traditional time-consuming internet search (e.g., on Google/Bing/Yahoo). [0430] Details of an intelligent and intuitive internet search interface have been described/disclosed (e.g., FIG. 1) in U.S. Non-Provisional patent Application No. U.S. Non-Provisional patent application Ser. No. 16/873,033 entitled “SYSTEM AND METHOD FOR MACHINE LEARNING AND AUGMENTED REALITY BASED USER APPLICATION”, filed on Jan. 18, 2020 and in its related U.S. non-provisional patent applications (with all benefit provisional patent applications) are incorporated in its entirety herein with this application.

[0431] In summary, the mobile internet appliance **300** can be coupled by a wireless network or a sensor network, [0432] wherein the mobile internet appliance **300** includes: [0433] (a) a system-on-chip, [0434] wherein the system-on-chip includes a microprocessor and/or a graphic processor, [0435] (b) a foldable display, [0436] (c) a radio transceiver or a sensor module, [0437] wherein the radio transceiver or the sensor module includes one or more electronic components, [0438] (d) a voice processing module or a voice processing algorithm; [0439] wherein the voice processing module includes one or more second electronic components, [0440] wherein the voice processing algorithm includes a first set of instructions to process a voice command, [0441] wherein the voice processing algorithm is stored in a first non-transitory storage media, [0442] wherein the mobile internet appliance **300** is further coupled with or further includes: [0443] (e) a natural language algorithm, and [0444] wherein the natural language algorithm includes a second set of instructions to understand the voice command in a natural spoken language of a user, [0445] wherein the natural language algorithm is stored in a second non-transitory storage media, [0446] (f) a learning algorithm or an intelligence algorithm, [0447] wherein the learning algorithm or the intelligence algorithm is based on or includes an artificial intelligence algorithm, [0448] wherein the learning algorithm or the intelligence algorithm includes a third set of instructions to provide learning or intelligence in response to an interest or a preference of the user, [0449] wherein the learning algorithm or the intelligence algorithm is stored in the second non-transitory storage media, [0450] wherein the first non-transitory storage media and the second non-transitory storage media are same or different, [0451] wherein the foldable display in (b), the radio transceiver or the sensor module in (c), the voice processing module or the voice processing algorithm in (d), the natural language algorithm in (e) and the learning algorithm or the intelligence algorithm in (f) are coupled with the system-on-chip in (a).

[0452] In summary, alternatively, the mobile internet appliance **300** can be coupled by a wireless network or a sensor network, [0453] wherein the mobile internet appliance **300** includes: [0454] (a) a system-on-chip, [0455] wherein the system-on-chip includes a microprocessor and/or a graphic processor, [0456] (b) a display including a solar cell (as illustrated in FIG. **5**D), [0457] (c) a radio transceiver or a sensor module, [0458] wherein the radio transceiver or the sensor module includes one or more electronic components, [0459] (d) a voice processing module or a voice processing algorithm; [0460] wherein the voice processing module includes one or more second electronic components, [0461] wherein the voice processing algorithm includes a first set of instructions to process a voice command, [0462] wherein the voice processing algorithm is stored in a first non-transitory storage media, [0463] wherein the mobile internet appliance **300** is further coupled with or further includes: [0464] (e) a natural language algorithm, and [0465] wherein the natural language algorithm includes a second set of instructions to understand the voice command in a natural spoken language of a user, [0466] wherein the natural language algorithm is stored in a second non-transitory storage media, [0467] (f) a learning algorithm or an intelligence algorithm,

[0468] wherein the learning algorithm or the intelligence algorithm is based on or includes an artificial intelligence algorithm, [0469] wherein the learning algorithm or the intelligence algorithm includes a third set of instructions to provide learning or intelligence in response to an interest or a preference of the user, [0470] wherein the learning algorithm or the intelligence algorithm is stored in the second non-transitory storage media, [0471] wherein the first non-transitory storage media and the second non-transitory storage media are same or different, [0472] wherein the foldable display in (b), the radio transceiver or the sensor module in (c), the voice processing module or the voice processing algorithm in (d), the natural language algorithm in (e) and the learning algorithm or the intelligence algorithm in (f) are coupled with the system-on-chip in (a).

[0473] Furthermore, the mobile internet appliance **300** can be coupled by or includes a neural processor or a photonic learning neural processor.

[0474] Details of a photonic neural learning processor have been described/disclosed in U.S. Non-Provisional patent application Ser. No. 16/602,404 entitled "SYSTEM AND METHOD OF AMBIENT/PERVASIVE USER/HEALTHCARE EXPERIENCE", filed on Sep. 28, 2019 and its related priority patent applications.

[0475] Details of a mobile internet appliance **300** have been described/disclosed in U.S. Non-Provisional patent application Ser. No. 11/952,001, entitled "Dynamic Intelligent Bidirectional Optical and Wireless Access Communication System, filed on Dec. 6, 2007, (which resulted in a U.S. Pat. No. 8,073,331, issued on Dec. 6, 2011), which claims the benefit of priority to [0476] U.S. Provisional patent application Ser. No. 60/970,487 entitled "Intelligent Internet Device", filed on Sep. 6, 2007, [0477] U.S. Provisional patent application Ser. No. 60/883,727 entitled "Wavelength Shifted Dynamic Bidirectional System", filed on Jan. 5, 2007, [0478] U.S. Provisional patent application Ser. No. 60/868,838 entitled "Wavelength Shifted Dynamic Bidirectional System", filed on Dec. 6, 2006.

Example C: Integration & Utilization Of Intelligent (Self-Learning) Subsystem

[0479] Without compromising the spirit of this invention, in another embodiment, the social wallet electronic module **280** and/or mobile internet device **300** (as described in this specification) can be replaced by an intelligent (self-learning) subsystem.

[0480] Details of an intelligent (self-learning) subsystem have been described/disclosed in U.S. Non-Provisional patent application Ser. No. 18/831,206 entitled "Intelligent (Self-Learning) Subsystem in Access Networks, filed on Sep. 14, 2024.

[0481] The entire contents of in U.S. Non-Provisional patent application Ser. No. 18/831,206 entitled "Intelligent (Self-Learning) Subsystem in Access Networks, filed on Sep. 14, 2024 is hereby incorporated by reference, as if they are reproduced herein in their entirety.

[0482] In general, but not limited to an intelligent (self-learning) subsystem is described/disclosed in U.S. Non-Provisional patent application Ser. No. 18/831,206 entitled "Intelligent (Self-Learning) Subsystem in Access Networks, filed on Sep. 14, 2024 as: [0483] An intelligent subsystem, which is self-learning intelligent subsystem, [0484] wherein the intelligent subsystem is operable with a wireless network or Zigbee, [0485] wherein the intelligent subsystem is sensor-aware and/or context-aware, [0486] wherein the intelligent subsystem includes [0487] (a) a radio transceiver, [0488] wherein the radio transceiver includes one or more first electronic components, [0489] (b) a microphone, [0490] (c) a voice processing module to process an audio signal, [0491] wherein the voice processing module includes one or more second electronic components, [0492] wherein the audio signal is a voice command, [0493] (d) a biometric sensor, [0494] (e) a location measurement module, [0495] wherein the location measurement module is selected from a group consisting of a radio frequency identification module, a Bluetooth module, a WiFi module, a global positioning system module and an indoor positioning system module, [0496] (f) a near-field component and [0497] (g) a System on Chip and/or Super system on Chip, [0498] wherein the System on Chip includes: [0499] one or more (i) processor-specific electronic integrated circuits (EICs) and (ii) first sets of computer implementable instructions, [0500] wherein at least one of the one or more processor-specific electronic integrated circuits includes one or more central processors, [0501] (A sequential workload can be executed by a central processor. But, when there is a parallel workload, then the central processor can assign to a parallel processor, which (a parallel processor) can be optimized for (i) Tolerating memory latency-meaning when a first thread calls to memory, a second thread can start running; (ii) Sufficient and flexible communication between chains of processing instructions that are running in parallel; (iii) Efficient synchronization (utilizing a protocol/algorithm allowing any functional unit can communicate to any other one, as needed) enabling execution of the parallel parts of a code in correct order; and (iv) The ability to use the functional units that perform both mathematical and logical operations simultaneously.) [0502] wherein the System on Chip has one or more multipliers of matrices, [0503] wherein at least one of the one or more first sets of computer implementable instructions is an embedded computer implementable instructions and processed on the System on Chip, [0504] Furthermore, alternatively, a System on Chip can be as follows: [0505] wherein the System-on-a-Chip includes one or more (i) central processors and (ii) a sets of computer implementable instructions, [0506] wherein at least one of the one or more sets of computer implementable instructions includes embedded computer implementable instructions and is processed on the System-on-a-Chip, [0507] wherein at least one of the one or more central processors includes one or more (i) processing cores and (ii) memory units, [0508] wherein at least one of the one or more processing cores and at least one of the one or more memory units are coupled in an intertwined pattern in a two-dimensional (2-D) arrangement or a three-dimensional (3-D) arrangement, [0509] wherein at least one of the one or more memory units includes one or more memory elements, wherein at least one of the one or more memory elements includes one or more memory circuits, [0510] wherein Super system on Chip includes memristors, [0511] wherein the intelligent subsystem is communicatively interfaced with: [0512] (i) a second set of computer implementable instructions to process the audio signal, [0513] (ii) a third set of computer implementable instructions to provide a recommendation inferred from an interest or a preference, that was received at the intelligent subsystem, wherein the third set of computer implementable instructions includes natural language processing, [0514] (iii) a fourth set of computer implementable instructions in artificial neural networks (ANN), [0515] wherein the artificial neural networks include a transformer model based computer implementable instructions or a diffusion model based computer implementable instructions and [0516] (iv) a fifth set of computer implementable instructions to analyze and interpret contextual data, [0517] wherein the second set of computer implementable instructions, the third set of computer implementable instructions, the fourth set of computer implementable instructions and the fifth set of computer implementable instructions are stored in one or more non-transitory storage media, located either locally on the intelligent subsystem or in a cloud server.

[0518] Furthermore, the fourth set of computer implementable instructions can include an evolutionary algorithm based computer implementable instructions or a game theory algorithm based computer implementable instructions.

[0519] Furthermore, the fourth set of computer implementable instructions can include Kolmogorov-Arnold based neural network architecture (KANN) or Poisson flow generative model++ (PFGM++).

[0520] The intelligent subsystem can be communicatively interfaced with a self-learning chatbot, wherein the self-learning chatbot may include a set of computer implementable instructions in natural language processing, stored in the one or more non-transitory storage media, located either locally on the intelligent subsystem or in the cloud server.

[0521] The intelligent subsystem can be communicatively interfaced with an autonomous artificial intelligence agent.

[0522] Alternatively, as an example, a computer implemented method can be described including:

[0523] (a) accessing, by an intelligent subsystem of a first user, via a wired network or a wireless network, a web portal enabled by a learning or relearning classical computer, a learning or relearning optical computer, or a learning or relearning quantum computer, [0524] wherein the web portal can include at least a first user profile associated with the first user and a second user profile associated with a second user, [0525] wherein the learning or relearning classical computer can include one or more cloud computers, premise computers, or mobile computers, wherein the learning or relearning classical computer can include one or more processors or one or more first neural network based processors, executing one or more computer implementable instructions and machine learning algorithms, on one or more non-transitory storage media to implement the web portal, [0526] wherein the learning or relearning optical computer can include one or more photonic neural learning processors, wherein the one photonic neural learning processor can include (i) memristors, (ii) one or more optical waveguides, or (iii) one or more optical emitters systems, wherein the one or more optical emitter systems are activated by weighted optical signals or weighted electrical signals, executing one or more computer implementable instructions and machine learning algorithms, on one or more non-transitory storage media to implement the web portal, [0527] wherein the learning or relearning quantum computer can include one or more quantum bits (qubits) executing one or more quantum computer enhanced algorithms or one or more machine learning algorithms to implement the web portal, [0528] wherein the intelligent subsystem can be operable with the wireless network or Zigbee, [0529] wherein the digital personal assistant is communicatively interfaced with a chatbot interface (including artificial intelligence based chatbot interface), [0530] wherein the digital personal assistant is configured to learn, adapt and take one or more autonomous actions to solve one or more problems on a user's behalf. [0531] wherein the intelligent subsystem can be sensor-aware and/or context-aware, [0532] wherein the intelligent subsystem can be self-learning, [0533] wherein the intelligent subsystem can include a digital personal assistant and/or an automated (artificial intelligence based) software agent, [0534] wherein the intelligent subsystem can further include: [0535] (i) a biometric sensor, [0536] (ii) a location measurement module, [0537] wherein the location measurement module can include one or more electronic components, [0538] wherein the location measurement module can be selected a radio frequency identification module, a Bluetooth module, a WiFi module, a global positioning system module or an indoor positioning system module, [0539] (iii) a near-field component, and [0540] (iv) a System on Chip, [0541] wherein the System on Chip can include: [0542] one or more (i) processor-specific electronic integrated circuits and (ii) first sets of computer implementable instructions, [0543] wherein at least one of the one or more processor-specific electronic integrated circuits can include one or more central processors, [0544] wherein the System on Chip has one or more multipliers of matrices, [0545] wherein at least one of the one or more first sets of computer implementable instructions has embedded computer implementable instructions and processed on the System on Chip [0546] wherein the intelligent subsystem can be communicatively interfaced with: [0547] (v) a second set of computer implementable instructions to process the audio signal, [0548] (vi) a third set of computer implementable instructions to provide a recommendation inferred from an interest or a preference, that was received at the intelligent subsystem, wherein the third set of computer implementable instructions can include natural language processing, [0549] (vii) a fourth set of computer implementable instructions in artificial neural networks, [0550] wherein the artificial neural networks include a transformer model based computer implementable instructions or a diffusion model based computer implementable instructions and [0551] (viii) a fifth set of computer implementable instructions to analyze and interpret contextual data, [0552] wherein the second set of computer implementable instructions, the third set of computer implementable instructions, the fourth set of computer implementable instructions and the fifth set of computer implementable instructions are stored in one or more non-transitory storage media, located either locally on the intelligent subsystem or in a cloud server, [0553] wherein said accessing the web portal can include: [0554] obtaining a first biometric scan of the first user from the biometric sensor of the first user, storing the first biometric scan of the first user, [0555] obtaining a second biometric scan of the first user from the biometric sensor of the first user,

[0556] wherein the second biometric scan of the first user can be a current biometric scan of the first user, comparing the first biometric scan of the first user with the second biometric scan of the first user to authenticate the first user, [0557] (b) in response to at least (a), automatically determining, by the web portal, that the first user is interested in purchasing a product or a service, [0558] wherein the product or the service may be linked with augmented reality (AR), [0559] (c) in response to at least (a) and (b), automatically determining, by the web portal, a near real time location of the first user, [0560] wherein the near real time location of the first user can be detected by the location measurement module of the intelligent subsystem of the first user, [0561] (d) in response to at least (a), (b) and (c), automatically querying, by the web portal, queried sellers (e.g., distant shopping centers) offering to sell the product or the service, [0562] (e) if needed, in response to at least (a), (b), (c) and (d), automatically selecting, by the web portal, one of the queried sellers, as a selected seller of the product or the service to purchase the product or the service, [0563] (f) in response to at least (a), (b), (c), (d) and (e), automatically connecting, by the web portal, the first user with the selected seller, [0564] (g) in response to at least (a), (b), (c), (d), (e) and (f), automatically forwarding, by the web portal, to the intelligent subsystem of the first user, in near real time, one or more sale offers to purchase the product or the service, from the selected seller, [0565] (h) in response to at least (a), (b), (c), (d), (e), (f) and (g), automatically accepting, by the web portal, like votes and dislike votes for the selected seller from the first user, the second user and a plurality of third users, [0566] (i) in response to at least (a), (b), (c), (d), (e), (f), (g) and (h), automatically determining a number of the like votes and a number of the dislike votes, [0567] (j) in response to at least (a), (b), (c), (d), (e), (f), (g), (h) and (i), automatically determining a seller score for the selected seller based on the number of the like votes and the number of the dislike votes by an algebraic equation, a statistical method, a statistical method coupled with conjoint analysis, or an algorithm based on quorum sensing, [0568] (k) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i) and (j), automatically displaying the seller score for the selected seller, and [0569] (l) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i), (j) and (k), performing at least one of (i) negotiating, by the selected seller with the first user, a price for the product or service or (ii) picking, by the first user, the price for the product or service (it should be noted that such negotiation can be done with the assistant of an automated (artificial intelligence based) software agent), [0570] (m) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i), (j), (k) and (l), the first user paying for the product or the service via the intelligent subsystem of the first user, wherein the said method steps in (a), (b), (c), (d), (e), (f), (g), (h), (i), (j), (k), (l) and (m) are at least an ordered combination or in an ordered sequence.

[0571] In the above method, the web portal can be coupled with one or more automated agents/(artificial intelligence based) software agents/bots/multimodal bots.

[0572] In the above method, the product, the service, or a service contract can be coupled with a blockchain.

[0573] In the above method, the web portal can receive an input data from a second user, a near-field communication (NFC) tag, a quick response (QR) code or an object, wherein the object can include a sensor and/or a wireless transmitter.

[0574] In the above method, the first user paying for the product or the service by transferring a currency or a check (or even a bitcoin) from the first user to the selected seller, to the second user, or to the plurality of third users by the web portal.

[0575] FIGS. **24**A-**24**H illustrate an application of the intelligent (self-learning) subsystem and a live view system for remote shopping.

[0576] FIG. **24**A illustrates a distant shopping mall (e.g., a distant shopping mall in Burj Khalifa in Dubai, UAE).

[0577] FIG. **24**B illustrates a user accesses to the above distant shopping mall via authentication on a biometric sensor (e.g., fingerprint/facial recognition/voice recognition) of the intelligent (self-learning) subsystem of the user. The intelligent (self-learning) subsystem can be a mobile based,

premise based or wearable.
[0578] In the case of the wearable intelligent (self-learning) subsystem, a projector, a time-of-flight light detection and ranging (LiDAR) device/computational camera and a microprocessor (e.g., a learning microprocessor) can be utilized to (i) project any relevant information (e.g., a text/picture) on a palm of a hand or (ii) enable aa palm of a hand as a virtual touch screen (keyboard). This is illustrated in FIG. **24**C.
[0579] Example of a text message on a palm of a hand is illustrated in FIG. **24**D, utilizing a projector, a time-of-flight light detection and ranging (LiDAR) device/computational camera and a microprocessor.
[0580] Various embodiments of light detection and ranging (LiDAR) devices and/or computational cameras have been described/disclosed in U.S. Non-Provisional patent application Ser. No. 17/803,388 entitled “SUPER SYSTEM ON CHIP”, filed on Jun. 15, 2022 (which resulted in a U.S. Pat. No. 11,892,746, issued on Feb. 6, 2024).
[0581] FIG. **24**E illustrates a block diagram of a system coupled with (i) a live shopping assistant or a virtual shopping assistant (including a virtual holographic shopping assistant), (ii) a live view system with augmented reality application (AR app) and (iii) the intelligent (self-learning) subsystem.
[0582] Generally the live view system with augmented reality application (AR app) can include all necessary hardware components and computer implementable instructions to enable an augmented reality application (AR app).
[0583] Utilizing the above live view system coupled with (i) the live/virtual shopping assistant (including a holographic shopping assistant) and (ii) the intelligent (self-learning) subsystem, a distant user can (i) search/access, (ii) virtually try products/services and (iii) securely pay for various products (e.g., dresses)/services (e.g., beauty makeup) and these are illustrated in FIG. **24**F and FIG. **24**G.
[0584] FIG. **24**H illustrates a method of distant shopping. In step **5000**, a user authenticates via a biometric sensor (e.g., fingerprint/facial recognition/voice recognition) of the intelligent (self-learning) subsystem. In step **5020**, the user connects to the live view system with augmented reality application (AR app). In step **5040**, the user virtually previews products/services. In step **5060**, the user consults with a live/virtual shopping assistant (including a virtual holographic shopping assistant). In step **5080**, the user virtually tries the products/services. In step **6000**, the live view system recommends other products/services to the user. In step **6020**, the user again authenticates via the biometric sensor (e.g., fingerprint/facial recognition/voice recognition) of the intelligent (self-learning) subsystem, if required prior to any payment. In step **6040**, the user pays (via the intelligent (self-learning) subsystem) for the products/services to be shipped to the user's address.
[0585] Furthermore, the method of distant shopping can be coupled with the web portal including a user's profile, as discussed in the previous paragraphs.
[0586] FIGS. **25**A-**25**E illustrate an application of the intelligent (self-learning) subsystem and the live view system with augmented reality application (AR app) for sharing a live game (e.g., a live basketball game).
[0587] FIG. **25**A illustrates a live basketball game.
[0588] In the case of the wearable intelligent (self-learning) subsystem, a projector, a time-of-flight light detection and ranging (LiDAR) device/computational camera and a microprocessor (e.g., a learning microprocessor) can be utilized to (i) project any relevant information (e.g., a text/picture) on a palm of a hand or (ii) enable a palm of a hand as a virtual touch screen (keyboard). This is illustrated in FIG. **25**B. Furthermore, a live user watches the live basketball game via the wearable intelligent (self-learning) subsystem and this is illustrated in FIG. **25**B.
[0589] Furthermore, the live user can review a key player's statistics on the palm of the live user and this is illustrated in FIG. **25**C, utilizing a projector, a time-of-flight light detection and ranging (LiDAR) device/computational camera and a microprocessor.

[0590] FIG. **25**D illustrates a block diagram of a system coupled with (i) the live view system with augmented reality application (AR app) and (ii) the intelligent (self-learning) subsystem. In FIG. **25**D the live user can be coupled with the live view system with augmented reality application (AR app) via the intelligent (self-learning) subsystem and similarly other remote viewers can be coupled with the live view system with augmented reality application (AR app) with via the intelligent (self-learning) subsystem.

[0591] FIG. **25**E illustrates a method of co-watching a live basketball game with the live user's live statistics and live comments. In step **7000**, the live user authenticates via a biometric sensor (e.g., fingerprint/facial recognition/voice recognition) of the intelligent (self-learning) subsystem. In step **7020**, the live user connects to the live view system with augmented reality application (AR app) via the intelligent (self-learning) subsystem. In step **7040**, remote users authenticate via biometric sensors (e.g., fingerprint/facial recognition/voice recognition) and connect to the live view system with augmented reality application (AR app). In step **7060**, the remote users pay for the live view system (e.g., similar to pay per view). In step **7080**, the live user shares live statistics and live comments with the remote users via the augmented reality application on the live view system.

[0592] Alternatively, as an example, a computer implemented method can be described including:

[0593] (a) accessing, by an intelligent subsystem of a first user, via a wired network or a wireless network, a web portal enabled by a learning or relearning classical computer, a learning or relearning optical computer, or a learning or relearning quantum computer, wherein the web portal can include at least a first user profile associated with the first user and a second user profile associated with a second user, [0594] wherein the learning or relearning classical computer can include one or more cloud computers, premise computers, or mobile computers, wherein the learning or relearning classical computer can include one or more processors or one or more first neural network based processors, executing one or more computer implementable instructions and machine learning algorithms, on one or more non-transitory storage media to implement the web portal, [0595] wherein the learning or relearning optical computer can include one or more photonic neural learning processors, wherein the one photonic neural learning processor can include (i) memristors, (ii) one or more optical waveguides, or (iii) one or more optical emitters systems, wherein the one or more optical emitter systems are activated by weighted optical signals or weighted electrical signals, executing one or more computer implementable instructions and machine learning algorithms, on one or more non-transitory storage media to implement the web portal, [0596] wherein the learning or relearning quantum computer can include one or more quantum bits (qubits) executing one or more quantum computer enhanced algorithms or one or more machine learning algorithms to implement the web portal, [0597] wherein the intelligent subsystem can be operable with the wireless network or Zigbee, [0598] wherein the digital personal assistant is communicatively interfaced with a chatbot interface (including artificial intelligence based chatbot interface), [0599] wherein the digital personal assistant is configured to learn, adapt and take one or more autonomous actions to solve one or more problems on a user's behalf, [0600] wherein the intelligent subsystem can be sensor-aware and/or context-aware, [0601] wherein the intelligent subsystem can be self-learning, [0602] wherein the intelligent subsystem can include a digital personal assistant and/or an automated (artificial intelligence based) software agent, [0603] wherein the intelligent subsystem can further include: [0604] (i) a biometric sensor, [0605] (ii) a location measurement module, [0606] wherein the location measurement module can include one or more electronic components, [0607] wherein the location measurement module can be selected a radio frequency identification module, a Bluetooth module, a WiFi module, a global positioning system module or an indoor positioning system module, [0608] (iii) a near-field component, and [0609] (iv) a System on Chip, [0610] wherein the System on Chip can include: [0611] one or more (i) processor-specific electronic integrated circuits and (ii) first sets of computer implementable instructions, [0612] wherein at least one of the one or more processor-specific electronic integrated circuits can include one or more central processors, [0613] wherein the System on Chip has one or more multipliers of matrices, [0614] wherein at least one of the one or more first sets of computer implementable instructions has embedded computer implementable instructions and processed on the System on Chip, [0615] wherein the intelligent subsystem can be communicatively interfaced with: [0616] (v) a second set of computer implementable instructions to process the audio signal, [0617] (vi) a third set of computer implementable instructions to provide a recommendation inferred from an interest or a preference, that was received at the intelligent subsystem, wherein the third set of computer implementable instructions can include natural language processing, [0618] (vii) a fourth set of computer implementable instructions in artificial neural networks, [0619] wherein the artificial neural networks include a transformer model based computer implementable instructions or a diffusion model based computer implementable instructions and [0620] (viii) a fifth set of computer implementable instructions to analyze and interpret contextual data, [0621] wherein the second set of computer implementable instructions, the third set of computer implementable instructions, the fourth set of computer implementable instructions and the fifth set of computer implementable instructions are stored in one or more non-transitory storage media, located either locally on the intelligent subsystem or in a cloud server, [0622] wherein said accessing the web portal can include: [0623] obtaining a first biometric scan of the first user from the biometric sensor of the first user, [0624] storing the first biometric scan of the first user, [0625] obtaining a second biometric scan of the first user from the biometric sensor of the first user, [0626] wherein the second biometric scan of the first user can be a current biometric scan of the first user, [0627] comparing the first biometric scan of the first user with the second biometric scan of the first user to authenticate the first user, [0628] (b) in response to at least (a), automatically determining, by the web portal, that the first user is interested in purchasing a product or a service, wherein the product or the service may be linked with augmented reality (AR), [0629] (c) in response to at least (a) and (b), the first user is virtually trying the product or the service with an augmented reality application (app), [0630] (d) in response to at least (a) (b) and (c), automatically determining, by the web portal, a near real time location of the first user, [0631] wherein the near real time location of the first user can be detected by the location measurement module of the intelligent subsystem of the first user, [0632] (e) in response to at least (a), (b), (c) and (d), automatically querying, by the web portal, queried sellers (e.g., distant shopping centers) offering to sell the product or the service, [0633] (f) if needed, in response to at least (a), (b), (c), (d) and (e), automatically selecting, by the web portal, one of the queried sellers, as a selected seller of the product or the service to purchase the product or the service, [0634] (g) in response to at least (a), (b), (c), (d), (e) and (f), automatically connecting, by the web portal, the first user with the selected seller, [0635] (h) in response to at least (a), (b), (c), (d), (e), (f) and (g), automatically forwarding, by the web portal, to the intelligent subsystem of the first user, in near real time, one or more sale offers to purchase the product or the service, from the selected seller, [0636] (i) in response to at least (a), (b), (c), (d), (e), (f), (g) and (h), automatically accepting, by the web portal, like votes and dislike votes for the selected seller from the first user, the second user and a plurality of third users, [0637] (j) in response to at least (a), (b), (c), (d), (e), (f), (g), (h) and (i), automatically determining a number of the like votes and a number of the dislike votes, [0638] (k) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i) and (j), automatically determining a seller score for the selected seller based on the number of the like votes and the number of the dislike votes by an algebraic equation, a statistical method, a statistical method coupled with conjoint analysis, or an algorithm based on quorum sensing, [0639] (l) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i), (j) and (k) automatically displaying the seller score for the selected seller, and [0640] (m) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i), (j), (k) and (l), performing at least one of (i) negotiating, by the selected seller with the first user, a price for the product or service or (ii) picking, by the first user, the price for the product or service, [0641] (n) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i), (j), (k), (l) and (m) the first user paying for the product or the service via the intelligent subsystem of the first user, wherein the said method steps in (a), (b), (c), (d), (e), (f), (g), (h), (i), (j), (k), (l), (m) and (n) are at least an ordered combination or in an ordered sequence.

[0642] Alternatively, as an example, a computer implemented method can be described including: [0643] (a) accessing, by an intelligent subsystem of a first user, via a wired network or a wireless network, a web portal enabled by a learning or relearning classical computer, a learning or relearning optical computer, or a learning or relearning quantum computer, [0644] wherein the web portal can include at least a first user profile associated with the first user and a second user profile associated with a second user, [0645] wherein the learning or relearning classical computer can include one or more cloud computers, premise computers, or mobile computers, wherein the learning or relearning classical computer can include one or more processors or one or more first neural network based processors, executing one or more computer implementable instructions and machine learning algorithms, on one or more non-transitory storage media to implement the web portal, [0646] wherein the learning or relearning optical computer can include one or more photonic neural learning processors, wherein the one photonic neural learning processor can include (i) memristors, (ii) one or more optical waveguides, or (iii) one or more optical emitters systems, wherein the one or more optical emitter systems are activated by weighted optical signals or weighted electrical signals, executing one or more computer implementable instructions and machine learning algorithms, on one or more non-transitory storage media to implement the web portal, [0647] wherein the learning or relearning quantum computer can include one or more quantum bits (qubits) executing one or more quantum computer enhanced algorithms or one or more machine learning algorithms to implement the web portal, [0648] wherein the intelligent subsystem can be operable with the wireless network or Zigbee, [0649] wherein the digital personal assistant is communicatively interfaced with a chatbot interface (including artificial intelligence based chatbot interface), [0650] wherein the digital personal assistant is configured to learn, adapt and take one or more autonomous actions to solve one or more problems on a user's behalf. [0651] wherein the intelligent subsystem can be sensor-aware and/or context-aware, [0652] wherein the intelligent subsystem can be self-learning, [0653] wherein the intelligent subsystem can include a digital personal assistant and/or an automated (artificial intelligence based) software agent, [0654] wherein the intelligent subsystem can further include: [0655] (i) a biometric sensor, [0656] (ii) a location measurement module, [0657] wherein the location measurement module can include one or more electronic components, [0658] wherein the location measurement module can be selected a radio frequency identification module, a Bluetooth module, a WiFi module, a global positioning system module or an indoor positioning system module, [0659] (iii) a near-field component, and [0660] (iv) a System on Chip, [0661] wherein the System on Chip can include: [0662] one or more (i) processor-specific electronic integrated circuits and (ii) first sets of computer implementable instructions, [0663] wherein at least one of the one or more processor-specific electronic integrated circuits can include one or more central processors, [0664] wherein the System on Chip has one or more multipliers of matrices, [0665] wherein at least one of the one or more first sets of computer implementable instructions has embedded computer implementable instructions and processed on the System on Chip [0666] wherein the intelligent subsystem can be communicatively interfaced with: [0667] (v) a second set of computer implementable instructions to process the audio signal, [0668] (vi) a third set of computer implementable instructions to provide a recommendation inferred from an interest or a preference, that was received at the intelligent subsystem, wherein the third set of computer implementable instructions can include natural language processing, [0669] (vii) a fourth set of computer implementable instructions in artificial neural networks, [0670] wherein the artificial neural networks include a transformer model based computer implementable instructions or a diffusion model based computer implementable instructions and [0671] (viii) a fifth set of computer implementable instructions to analyze and interpret contextual data, [0672] wherein the second set of computer implementable instructions, the third set of computer implementable instructions, the fourth set of computer implementable instructions and the fifth set of computer implementable instructions are stored in one or more non-transitory storage media, located either locally on the intelligent subsystem or in a cloud server, [0673] wherein said accessing the web portal can include: [0674] obtaining a first biometric scan of the first user from the biometric sensor of the first user, storing the first biometric scan of the first user, [0675] obtaining a second biometric scan of the first user from the biometric sensor of the first user, [0676] wherein the second biometric scan of the first user can be a current biometric scan of the first user, [0677] comparing the first biometric scan of the first user with the second biometric scan of the first user to authenticate the first user, [0678] (b) in response to at least (a), automatically determining, by the web portal, that the first user is interested in watching a live game, [0679] (c) in response to at least (a) and (b), the first user is connecting with other remote users, [0680] (d) in response to at least (a), (b) and (c), automatically forwarding, by the web portal, to the intelligent subsystem of the first user, in near real time, one or more sale offers to purchase a live game, [0681] (e) in response to at least (a), (b), (c) and (d), automatically accepting, by the web portal, like votes and dislike votes for the live game from the first user, the second user and a plurality of third users, [0682] (f) in response to at least (a), (b), (c), (d) and (e), automatically determining a number of the like votes and a number of the dislike votes for the live game, [0683] (g) in response to at least (a), (b), (c), (d), (e) and (f), automatically determining the cumulative like votes and a cumulative dislike votes for the live game by an algebraic equation, a statistical method, a statistical method coupled with conjoint analysis, or an algorithm based on quorum sensing, [0684] (h) in response to at least (a), (b), (c), (d), (e), (f) and (g), automatically displaying the cumulative like votes or cumulative dislike votes for the live game, [0685] (i) in response to at least (a), (b), (c), (d), (e), (f), (g) and (h), the first user securely paying for the live game, [0686] (j) in response to at least (a), (b), (c), (d), (e), (f), (g), (h) and (i), the remote users securely paying for the live game, [0687] (k) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i) and (j) the first user is sharing live statistics and live comments of the live game with the remote users, wherein the said method steps in (a), (b), (c), (d), (e), (f), (g), (h), (i), (i) and (k) are at least an ordered combination or in an ordered sequence.

PREFERRED EMBODIMENTS & SCOPE OF THE INVENTION

[0688] As used in the above disclosed specifications, the above disclosed specifications “/” has been used to indicate an “or”.

[0689] As used in the above disclosed specifications and in the claims, the singular forms “a”, “an”, and “the” also include the plural forms, unless the context clearly dictates otherwise.

[0690] As used in the above disclosed specifications, the term “includes” means “comprises”. Also the term “including” means “comprising”.

[0691] As used in the above disclosed specifications, the term “couples” or “coupled” does not exclude the presence of an intermediate element(s) between the coupled items.

[0692] Any dimension in the above disclosed specifications is by way of an approximation only and not by way of any limitation.

[0693] As used in the above disclosed specifications, unless otherwise specified in the relevant paragraph(s), a nanoscaled dimension shall generally mean a dimension from about 1 nm to about 1000 nm.

[0694] As used in the above disclosed specifications, the word “microprocessor” is synonymous with the word “processor”.

[0695] As used in the above disclosed specifications, the word “unit” is synonymous with the word “media unit” or with the word “media”.

[0696] As used in the above disclosed specifications, the word cloud based storage unit is synonymous with a cloud based server.

[0697] As used in the above disclosed specifications, a hardware module/module is defined as an integration of critical electrical/optical/radio/sensor components and circuits (and algorithms, if needed) to achieve a desired property of a hardware module/module.

[0698] As used in the above disclosed specifications, an algorithm is defined as organized set of instructions to achieve a desired task.
[0699] As used in the above disclosed specifications, a software module is defined as a collection of consistent algorithms to achieve a desired task.
[0700] As used in the above disclosed specifications, real time means near real time in practice.
[0701] Any example in the above disclosed specifications is by way of an example only and not by way of any limitation. Having described and illustrated the principles of the disclosed technology with reference to the illustrated embodiments, it will be recognized that the illustrated embodiments can be modified in any arrangement and detail with departing from such principles. The technologies from any example can be combined in any arrangement with the technologies described in any one or more of the other examples. Alternatives specifically addressed in this application are merely exemplary and do not constitute all possible examples. Claimed invention is disclosed as one of several possibilities or as useful separately or in various combinations. See *Novozymes A/S* v. *DuPont Nutrition Biosciences APS*, 723 F3d 1336,1347.
[0702] The best mode requirement "requires an inventor(s) to disclose the best mode contemplated by him/her, as of the time he/she executes the application, of carrying out the invention." " . . . [T]he existence of a best mode is a purely subjective matter depending upon what the inventor(s) actually believed at the time the application was filed." See *Bayer AG* v. *Schein Pharmaceuticals, Inc*. The best mode requirement still exists under the America Invents Act (AIA). At the time of the invention, the inventor(s) described preferred best mode embodiments of the present invention. The sole purpose of the best mode requirement is to restrain the inventor(s) from applying for a patent, while at the same time concealing from the public preferred embodiments of their inventions, which they have in fact conceived. The best mode inquiry focuses on the inventor(s)′ state of mind at the time he/she filed the patent application, raising a subjective factual question. The specificity of disclosure required to comply with the best mode requirement must be determined by the knowledge of facts within the possession of the inventor(s) at the time of filing the patent application. See *Glaxo, Inc*. v. *Novopharm Ltd.,* 52 F.3d 1043, 1050 (Fed. Cir. 1995). The above disclosed specifications are the preferred best mode embodiments of the present invention. However, they are not intended to be limited only to the preferred best mode embodiments of the present invention.
[0703] Embodiment by definition is a manner in which an invention can be made or used or practiced or expressed. "A tangible form or representation of the invention" is an embodiment.
[0704] Numerous variations and/or modifications are possible within the scope of the present invention. Accordingly, the disclosed preferred best mode embodiments are to be construed as illustrative only. Those who are skilled in the art can make various variations and/or modifications without departing from the scope and spirit of this invention. It should be apparent that features of one embodiment can be combined with one or more features of another embodiment to form a plurality of embodiments. The inventor(s) of the present invention is not required to describe each and every conceivable and possible future embodiment in the preferred best mode embodiments of the present invention. See *SRI Int'l v. Matsushita Elec. Corp. of America,* 775F.2d 1107, 1121, 227U.S.P.Q. (BNA) 577, 585 (Fed. Cir. 1985) (enbanc).
[0705] The scope and spirit of this invention shall be defined by the claims and the equivalents of the claims only. The exclusive use of all variations and/or modifications within the scope of the claims is reserved. The general presumption is that claim terms should be interpreted using their plain and ordinary meaning without improperly importing a limitation from the specification into the claims. See *Continental Circuits LLC* v. *Intel Corp*. (Appeal Number 2018-1076, Fed. Cir. Feb. 8, 2019) and *Oxford Immunotec Ltd*. v. *Qiagen, Inc*. et al., Action No. 15-cv-13124-NMG. Unless a claim term is specifically defined in the preferred best mode embodiments, then a claim term has an ordinary meaning, as understood by a person with an ordinary skill in the art, at the time of the present invention. Plain claim language will not be narrowed unless the inventor(s) of the present

invention clearly and explicitly disclaims broader claim scope. See *Sumitomo Dainippon Pharma Co*. v. *Emcure Pharm. Ltd*., Case Nos. 17-1798; -1799; -1800 (Fed. Cir. Apr. 16, 2018) (Stoll, J). As noted long ago: "Specifications teach. Claims claim". See *Rexnord Corp*. v. *Laitram Corp.,* 274 F.3d 1336, 1344 (Fed. Cir. 2001). The rights of claims (and rights of the equivalents of the claims) under the Doctrine of Equivalents-meeting the "Triple Identity Test" (a) performing substantially the same function, (b) in substantially the same way and (c) yielding substantially the same result. See *Crown Packaging Tech., Inc*. v. *Rexam Beverage Can Co.,* 559 F.3d 1308, 1312 (Fed. Cir. 2009)) of the present invention are not narrowed or limited by the selective imports of the specifications (of the preferred embodiments of the present invention) into the claims.

[0706] While "absolute precision is unattainable" in patented claims, the definiteness requirement "mandates clarity." See *Nautilus, Inc*. v. *Biosig Instruments, Inc.,* 527 U.S., 134 S. Ct. 2120, 2129, 110 USPQ2d 1688, 1693 (2014). Definiteness of claim language must be analyzed NOT in a vacuum, but in light of: [0707] (a) The content of the particular application disclosure, [0708] (b) The teachings of any prior art and [0709] (c) The claim interpretation that would be given by one possessing the ordinary level of skill in the pertinent art at the time the invention was made. (Id.). See *Orthokinetics, Inc*. v. *Safety Travel Chairs, Inc.,* 806 F.2d 1565, 1 USPQ2d 1081 (Fed. Cir. 1986)

[0710] There are number of ways the written description requirement is satisfied. Applicant(s) does not need to describe every claim element exactly, because there is no such requirement (MPEP § 2163). Rather to satisfy the written description requirement, all that is required is "reasonable clarity" (MPEP § 2163.02). An adequate description may be made in any way through express, implicit or even inherent disclosures in the application, including word, structures, figures, diagrams and/or equations (MPEP §§ 2163(I), 2163.02). The set of claims in this invention generally covers a set of sufficient number of embodiments to conform to written description and enablement doctrine. See *Ariad Pharm., Inc*. v. *Eli Lilly* & *Co.,* 598 F.3d 1336, 1355 (Fed. Cir. 2010), *Regents of the University of California* v. *Eli Lilly* & *Co.,* 119 F.3d 1559 (Fed. Cir. 1997) & *Amgen Inc*. v. *Chugai Pharmaceutical Co.* 927 F.2d 1200 (Fed. Cir. 1991).

[0711] Furthermore, *Amgen Inc*. v. *Chugai Pharmaceutical Co*. exemplifies Federal Circuit's strict enablement requirements. Additionally, the set of claims in this invention is intended to inform the scope of this invention with "reasonable certainty". See *Interval Licensing, LLC* v. *AOL Inc*. (Fed. Cir. Sep. 10, 2014). A key aspect of the enablement requirement is that it only requires that others will not have to perform "undue experimentation" to reproduce it. Enablement is not precluded by the necessity of some experimentation, "[t]he key word is 'undue', not experimentation." Enablement is generally considered to be the most important factor for determining the scope of claim protection allowed. The scope of enablement must be commensurate with the scope of the claims. However, enablement does not require that an inventor disclose every possible embodiment of his invention. The scope of enablement must be commensurate with the scope of the claims. The scope of the claims must be less than or equal to the scope of enablement. See *Promega* v. *Life Technologies* Fed. Cir., December 2014, *Magsil* v. *Hitachi Global Storage* Fed. Cir. August 2012.

[0712] The term "means" was not used nor intended nor implied in the disclosed preferred best mode embodiments of the present invention. Thus, the inventor(s) has not limited the scope of the claims as mean plus function.

[0713] An apparatus claim with functional language is not an impermissible "hybrid" claim; instead, it is simply an apparatus claim including functional limitations. Additionally, "apparatus claims are not necessarily indefinite for using functional language . . . [f]unctional language may also be employed to limit the claims without using the means-plus-function format." See *National Presto Industries, Inc*. v. *The West Bend Co.,* 76 F. 3d 1185 (Fed. Cir. 1996), *R.A.C.C. Indus*. v. *Stun-Tech, Inc.,* 178 F.3d 1309 (Fed. Cir. 1998) (unpublished), *Microprocessor Enhancement Corp.* v. *Texas Instruments Inc*. & *Williamson* v. *Citrix Online, LLC,* 792 F.3d 1339 (2015).

Claims

1. A computer implemented method comprising: (a) accessing, by an intelligent subsystem of a first user, via a wired network or a wireless network, a web portal enabled by a learning or relearning classical computer, a learning or relearning optical computer, or a learning or relearning quantum computer, wherein the web portal comprises at least a first user profile associated with the first user and a second user profile associated with a second user, wherein the learning or relearning classical computer is one or more cloud computers, premise computers, or mobile computers, wherein the learning or relearning classical computer comprises one or more processors or one or more first neural network based processors, executing one or more computer implementable instructions and machine learning algorithms, on one or more non-transitory storage media to implement the web portal, wherein the learning or relearning optical computer comprises one or more photonic neural learning processors, wherein the one photonic neural learning processor includes (i) memristors, (ii) one or more optical waveguides, or (iii) one or more optical emitters systems, wherein the one or more optical emitter systems are activated by weighted optical signals or weighted electrical signals, executing one or more computer implementable instructions and machine learning algorithms, on one or more non-transitory storage media to implement the web portal, wherein the learning or relearning quantum computer comprises one or more quantum bits (qubits) executing one or more quantum computer enhanced algorithms or one or more machine learning algorithms to implement the web portal, wherein the intelligent subsystem is operable with the wireless network or Zigbee, wherein the intelligent subsystem is sensor-aware and/or context-aware, wherein the intelligent subsystem comprises: (i) a biometric sensor; and (ii) a system-on-chip (SoC), wherein the system-on-chip (SoC) includes: one or more (i) processor-specific electronic integrated circuits (EICs) and (ii) first sets of computer implementable instructions, wherein at least one of the one or more processor-specific electronic integrated circuits (EICs) comprises one or more central processors, wherein the system-on-chip (SoC) has one or more multipliers of matrices, wherein at least one of the one or more first sets of computer implementable instructions is embedded computer implementable instructions and is processed on the system-on-chip (SoC), wherein the intelligent subsystem is communicatively interfaced with: (iii) a second set of computer implementable instructions to provide a recommendation inferred from an interest or a preference, that was received at the intelligent subsystem, wherein the second set of computer implementable instructions comprises natural language processing (NLP), (iv) a third set of computer implementable instructions in artificial neural networks (ANN), wherein the artificial neural networks (ANN) include a transformer model based computer implementable instructions or a diffusion model based computer implementable instructions and (v) a fourth set of computer implementable instructions to analyze and interpret contextual data, wherein the second set of computer implementable instructions, the third set of computer implementable instructions and the fourth set of computer implementable instructions are stored in one or more non-transitory storage media, located either locally on the intelligent subsystem or in a cloud server, wherein said accessing the web portal comprises: obtaining a first biometric scan of the first user from the biometric sensor of the first user, storing the first biometric scan of the first user, obtaining a second biometric scan of the first user from the biometric sensor of the first user, wherein the second biometric scan of the first user is a current biometric scan of the first user, comparing the first biometric scan of the first user with the second biometric scan of the first user to authenticate the first user; (b) in response to at least (a), listing or linking, by the first user, a product or a service for purchase on the first user profile in the web portal, (c) in response to at least (a) and (b), automatically determining, by the web portal, that the first user is interested in purchasing the product or the service; (d) in response to at least (a), (b) and (c), automatically determining, by the web portal, a near real time location of the first user; (e) in response to at least (a), (b), (c) and (d),

automatically querying, by the web portal, queried sellers offering to sell the product or the service; (f) in response to at least (a), (b), (c), (d) and (e), automatically selecting, by the web portal, one of the queried sellers, as a selected seller of the product or the service to purchase the product or the service; (g) in response to at least (a), (b), (c), (d), (e) and (f), automatically connecting, by the web portal, the first user with the selected seller; (h) in response to at least (a), (b), (c), (d), (e), (f) and (g), automatically forwarding, by the web portal, to the intelligent subsystem of the first user, in near real time, one or more sale offers to purchase the product or the service, from the selected seller; (i) in response to at least (a), (b), (c), (d), (e), (f), (g) and (h), automatically accepting, by the web portal, like votes and dislike votes for the selected seller from the first user, the second user and a plurality of third users; (j) in response to at least (a), (b), (c), (d), (e), (f), (g), (h) and (i), automatically determining a number of the like votes and a number of the dislike votes; (k) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i) and (j), automatically determining a seller score for the selected seller based on the number of the like votes and the number of the dislike votes by an algebraic equation, a statistical method, a statistical method coupled with conjoint analysis, or an algorithm based on quorum sensing; (l) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i), (j) and (k), automatically displaying the seller score for the selected seller; and (m) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i), (j), (k) and (l), performing at least one of (i) negotiating, by the selected seller with the first user, a price for the product or the service or (ii) picking, by the first user, the price for the product or the service.
2. The method according to claim 1, wherein the web portal is coupled with one or more software agents or bots or multimodal bots, wherein at least one of the multimodal bots comprises one or more computer implementable interfaces to comprehend and react to a voice, a text and a visual input.
3. The method according to claim 1, wherein the product, the service, or a service contract is coupled with a blockchain.
3. The method according to claim 1, wherein the web portal is receiving an input data from the second user, a near-field communication (NFC) tag, a quick response (QR) code, or an object, wherein the object comprises a sensor and/or a wireless transmitter.
4. The method according to claim 1, further comprising the first user paying for the product or the service by transferring a currency or a bitcoin from the first user to the selected seller, to the second user, or to the plurality of third users by the web portal.
5. A computer implemented method comprising: (a) accessing, by an intelligent subsystem of a first user, via a wired network or a wireless network, a web portal enabled by a learning or relearning classical computer, a learning or relearning optical computer, or a learning or relearning quantum computer, wherein the web portal comprises at least a first user profile associated with the first user and a second user profile associated with a second user, wherein the learning or relearning classical computer is one or more cloud computers, premise computers, or mobile computers, wherein the learning or relearning classical computer comprises one or more processors or one or more first neural network based processors, executing one or more computer implementable instructions and machine learning algorithms, on one or more non-transitory storage media to implement the web portal, wherein the learning or relearning optical computer comprises one or more photonic neural learning processors, wherein the one photonic neural learning processor includes (i) memristors, (ii) one or more optical waveguides, or (iii) one or more optical emitters systems, wherein the one or more optical emitter systems are activated by weighted optical signals or weighted electrical signals, executing one or more computer implementable instructions and machine learning algorithms, on one or more non-transitory storage media to implement the web portal, wherein the learning or relearning quantum computer comprises one or more quantum bits (qubits) executing one or more quantum computer enhanced algorithms or one or more machine learning algorithms to implement the web portal, wherein the intelligent subsystem is operable with the wireless network or Zigbee, wherein the intelligent subsystem is sensor-aware and/or context-aware,

wherein the intelligent subsystem comprises: (i) a biometric sensor; and (ii) a system-on-chip (SoC), wherein the system-on-chip (SoC) includes: one or more (i) processor-specific electronic integrated circuits (EICs) and (ii) first sets of computer implementable instructions, wherein at least one of the one or more processor-specific electronic integrated circuits (EICs) comprises one or more central processors, wherein the system-on-chip (SoC) has one or more multipliers of matrices, wherein at least one of the one or more first sets of computer implementable instructions is embedded computer implementable instructions and is processed on the system-on-chip (SoC), wherein the intelligent subsystem is communicatively interfaced with: (iii) a second set of computer implementable instructions to provide a recommendation inferred from an interest or a preference, that was received at the intelligent subsystem, wherein the second set of computer implementable instructions comprises natural language processing (NLP), (iv) a third set of computer implementable instructions in artificial neural networks (ANN), wherein the artificial neural networks (ANN) include a transformer model based computer implementable instructions or a diffusion model based computer implementable instructions and (v) a fourth set of computer implementable instructions to analyze and interpret contextual data, wherein the second set of computer implementable instructions, the third set of computer implementable instructions and the fourth set of computer implementable instructions are stored in one or more non-transitory storage media, located either locally on the intelligent subsystem or in a cloud server, wherein said accessing the web portal comprises: obtaining a first biometric scan of the first user from the biometric sensor of the first user, storing the first biometric scan of the first user, obtaining a second biometric scan of the first user from the biometric sensor of the first user, wherein the second biometric scan of the first user is a current biometric scan of the first user, comparing the first biometric scan of the first user with the second biometric scan of the first user to authenticate the first user; (b) in response to at least (a), listing or linking, by the first user, a product or a service for purchase on the first user profile in the web portal, wherein the product or the service is linked with augmented reality (AR); (c) in response to at least (a) and (b), automatically determining, by the web portal, that the first user is interested in purchasing the product or the service; (d) in response to at least (a), (b) and (c), automatically determining, by the web portal, a near real time location of the first user; (e) in response to at least (a), (b), (c) and (d), automatically querying, by the web portal, queried sellers offering to sell the product or the service; (f) in response to at least (a), (b), (c), (d) and (e), automatically selecting, by the web portal, one of the queried sellers, as a selected seller of the product or the service to purchase the product or the service; (g) in response to at least (a), (b), (c), (d), (e) and (f), automatically connecting, by the web portal, the first user with the selected seller; (h) in response to at least (a), (b), (c), (d), (e), (f) and (g), automatically forwarding, by the web portal, to the intelligent subsystem of the first user, in near real time, one or more sale offers to purchase the product or the service, from the selected seller; (i) in response to at least (a), (b), (c), (d), (e), (f), (g) and (h), automatically accepting, by the web portal, like votes and dislike votes for the selected seller from the first user, the second user and a plurality of third users; (j) in response to at least (a), (b), (c), (d), (e), (f), (g), (h) and (i), automatically determining a number of the like votes and a number of the dislike votes; (k) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i) and (j), automatically determining a seller score for the selected seller based on the number of the like votes and the number of the dislike votes by an algebraic equation, a statistical method, a statistical method coupled with conjoint analysis, or an algorithm based on quorum sensing; (l) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i), (j) and (k), automatically displaying the seller score for the selected seller; and (m) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i), (j), (k) and (l), performing at least one of (i) negotiating, by the selected seller with the first user, a price for the product or the service or (ii) picking, by the first user, the price for the product or the service.

6. The method according to claim 5, wherein the web portal is coupled with one or more software agents or bots or multimodal bots, wherein at least one of the multimodal bots comprises one or more computer implementable interfaces to comprehend and react to a voice, a text and a visual input.

7. The method according to claim 5, wherein the product, the service, or a service contract is coupled with a blockchain.

8. The method according to claim 5, wherein the web portal is receiving an input data from the second user, a near-field communication (NFC) tag, a quick response (QR) code, or an object, wherein the object comprises a sensor and/or a wireless transmitter.

9. The method according to claim 5, further comprising the first user paying for the product or the service by transferring a currency or a bitcoin from the first user to the selected seller, to the second user, or to the plurality of third users by the web portal.

10. A computer implemented method comprising: (a) accessing, by an intelligent subsystem of a first user, via a wired network or a wireless network, a web portal enabled by a learning or relearning classical computer, a learning or relearning optical computer, or a learning or relearning quantum computer, wherein the web portal comprises at least a first user profile associated with the first user and a second user profile associated with a second user, wherein the learning or relearning classical computer is one or more cloud computers, premise computers, or mobile computers, wherein the learning or relearning classical computer comprises one or more processors or one or more first neural network based processors, executing one or more computer implementable instructions and machine learning algorithms, on one or more non-transitory storage media to implement the web portal, wherein the learning or relearning optical computer comprises one or more photonic neural learning processors, wherein the one photonic neural learning processor includes (i) memristors, (ii) one or more optical waveguides, or (iii) one or more optical emitters systems, wherein the one or more optical emitter systems are activated by weighted optical signals or weighted electrical signals, executing one or more computer implementable instructions and machine learning algorithms, on one or more non-transitory storage media to implement the web portal, wherein the learning or relearning quantum computer comprises one or more quantum bits (qubits) executing one or more quantum computer enhanced algorithms or one or more machine learning algorithms to implement the web portal, wherein the intelligent subsystem is operable with the wireless network or Zigbee, wherein the intelligent subsystem is sensor-aware and/or context-aware, wherein the intelligent subsystem is self-learning, wherein the intelligent subsystem comprises: (i) a biometric sensor; and (ii) a system-on-chip (SoC), wherein the system-on-chip (SoC) includes: one or more (i) processor-specific electronic integrated circuits (EICs), (ii) graphic processors and (ii) first sets of computer implementable instructions, wherein at least one of the one or more processor-specific electronic integrated circuits (EICs) comprises one or more central processors, wherein the system-on-chip (SoC) has one or more multipliers of matrices, wherein at least one of the one or more first sets of computer implementable instructions is embedded computer implementable instructions and is processed on the system-on-chip (SoC), wherein the intelligent subsystem is communicatively interfaced with: (iii) a second set of computer implementable instructions to provide a recommendation inferred from an interest or a preference, that was received at the intelligent subsystem, wherein the second set of computer implementable instructions comprises natural language processing (NLP), (iv) a third set of computer implementable instructions in artificial neural networks (ANN), wherein the artificial neural networks (ANN) include a transformer model based computer implementable instructions or a diffusion model based computer implementable instructions and (v) a fourth set of computer implementable instructions to analyze and interpret contextual data, wherein the second set of computer implementable instructions, the third set of computer implementable instructions and the fourth set of computer implementable instructions are stored in one or more non-transitory storage media, located either locally on the intelligent subsystem or in a cloud server, wherein said accessing the web portal comprises: obtaining a first biometric scan of the first user from the biometric sensor of the first user, storing the first biometric scan of the first user, obtaining a second biometric scan of the first user from the biometric sensor of the first user, wherein the second biometric scan of the first user is a current biometric scan of the first user, comparing the first biometric scan of the first user with the second biometric scan of the first user to authenticate the first user; (b) in response to at least (a), listing or linking, by the first user, a product or a service for purchase on the first user profile in the web portal, (c) in response to at least (a) and (b), automatically determining, by the web portal, that the first user is interested in purchasing the product or the service; (d) in response to at least (a), (b) and (c), automatically determining, by the web portal, a near real time location of the first user; (e) in response to at least (a), (b), (c) and (d), automatically querying, by the web portal, queried sellers offering to sell the product or the service; (f) in response to at least (a), (b), (c), (d) and (e), automatically selecting, by the web portal, one of the queried sellers, as a selected seller of the product or the service to purchase the product or the service; (g) in response to at least (a), (b), (c), (d), (e) and (f), automatically connecting, by the web portal, the first user with the selected seller; (h) in response to at least (a), (b), (c), (d), (e), (f) and (g), automatically forwarding, by the web portal, to the intelligent subsystem of the first user, in near real time, one or more sale offers to purchase the product or the service, from the selected seller; (i) in response to at least (a), (b), (c), (d), (e), (f), (g) and (h), automatically accepting, by the web portal, like votes and dislike votes for the selected seller from the first user, the second user and a plurality of third users; (j) in response to at least (a), (b), (c), (d), (e), (f), (g), (h) and (i), automatically determining a number of the like votes and a number of the dislike votes; (k) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i) and (j), automatically determining a seller score for the selected seller based on the number of the like votes and the number of the dislike votes by an algebraic equation, a statistical method, a statistical method coupled with conjoint analysis, or an algorithm based on quorum sensing; (l) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i), (j) and (k), automatically displaying the seller score for the selected seller; and (m) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i), (j), (k) and (l), performing at least one of (i) negotiating, by the selected seller with the first user, a price for the product or the service or (ii) picking, by the first user, the price for the product or the service, wherein the said method steps in (a), (b), (c), (d), (e), (f), (g), (h), (i), (j), (k), (l) and (m) are at least an ordered combination or in an ordered sequence.

11. The method according to claim 10, wherein the web portal is coupled with one or more software agents or bots or multimodal bots, wherein at least one of the multimodal bots comprises one or more computer implementable interfaces to comprehend and react to a voice, a text and a visual input.

12. The method according to claim 10, wherein the product, the service, or a service contract is coupled with a blockchain.

13. The method according to claim 10, wherein the web portal is receiving an input data from the second user, a near-field communication (NFC) tag, a quick response (QR) code, or an object, wherein the object comprises a sensor and/or a wireless transmitter.

14. The method according to claim 10, further comprising the first user paying for the product or the service by transferring a currency or a bitcoin from the first user to the selected seller, to the second user, or to the plurality of third users by the web portal.

15. A computer implemented method comprising: (a) accessing, by an intelligent subsystem of a first user, via a wired network or a wireless network, a web portal enabled by a learning or relearning classical computer, a learning or relearning optical computer, or a learning or relearning quantum computer, wherein the web portal comprises at least a first user profile associated with the first user and a second user profile associated with a second user, wherein the learning or relearning classical computer is one or more cloud computers, premise computers, or mobile computers, wherein the learning or relearning classical computer comprises one or more processors or one or more first neural network based processors, executing one or more computer implementable instructions and machine learning algorithms, on one or more non-transitory storage media to implement the web portal, wherein the learning or relearning optical computer comprises one or more photonic neural learning processors, wherein the one photonic neural learning processor includes (i) memristors, (ii) one or more optical waveguides, or (iii) one or more optical emitters systems, wherein the one or more optical emitter systems are activated by weighted optical signals or weighted electrical signals, executing one or more computer implementable instructions and machine learning algorithms, on one or more non-transitory storage media to implement the web portal, wherein the learning or relearning quantum computer comprises one or more quantum bits (qubits) executing one or more quantum computer enhanced algorithms or one or more machine learning algorithms to implement the web portal, wherein the intelligent subsystem is operable with the wireless network or Zigbee, wherein the intelligent subsystem is sensor-aware and/or context-aware, wherein the intelligent subsystem is self-learning, wherein the intelligent subsystem is communicatively interfaced with one or more bots, wherein the intelligent subsystem comprises: (i) a biometric sensor; and (ii) a system-on-chip (SoC), wherein the system-on-chip (SoC) includes: one or more (i) processor-specific electronic integrated circuits (EICs), (ii) graphic processors and (iii) first sets of computer implementable instructions, wherein at least one of the one or more processor-specific electronic integrated circuits (EICs) comprises one or more central processors, wherein the system-on-chip (SoC) includes one or more on-sensor processing circuits, wherein at least one of the one or more on-sensor processing circuit includes a digital signal processor (DSP), wherein the system-on-chip (SoC) has one or more multipliers of matrices, wherein at least one of the one or more first sets of computer implementable instructions is embedded computer implementable instructions and is processed on the system-on-chip (SoC), wherein the intelligent subsystem is communicatively interfaced with: (iii) a second set of computer implementable instructions to provide a recommendation inferred from an interest or a preference, that was received at the intelligent subsystem, wherein the second set of computer implementable instructions comprises natural language processing (NLP), (iv) a third set of computer implementable instructions in artificial neural networks (ANN), wherein the artificial neural networks (ANN) include a transformer model based computer implementable instructions or a diffusion model based computer implementable instructions and (v) a fourth set of computer implementable instructions to analyze and interpret contextual data, wherein the second set of computer implementable instructions, the third set of computer implementable instructions and the fourth set of computer implementable instructions are stored in one or more non-transitory storage media, located either locally on the intelligent subsystem or in a cloud server, wherein said accessing the web portal comprises: obtaining a first biometric scan of the first user from the biometric sensor of the first user, storing the first biometric scan of the first user, obtaining a second biometric scan of the first user from the biometric sensor of the first user, wherein the second biometric scan of the first user is a current biometric scan of the first user, comparing the first biometric scan of the first user with the second biometric scan of the first user to authenticate the first user; (b) in response to at least (a), listing or linking, by the first user, a product or a service for purchase on the first user profile in the web portal, (c) in response to at least (a) and (b), automatically determining, by the web portal, that the first user is interested in purchasing the product or the service; (d) in response to at least (a), (b) and (c), automatically determining, by the web portal, a near real time location of the first user; (e) in response to at least (a), (b), (c) and (d), automatically querying, by the web portal, queried sellers offering to sell the product or the service; (f) in response to at least (a), (b), (c), (d) and (e), automatically selecting, by the web portal, one of the queried sellers, as a selected seller of the product or the service to purchase the product or the service; (g) in response to at least (a), (b), (c), (d), (e) and (f), automatically connecting, by the web portal, the first user with the selected seller; (h) in response to at least (a), (b), (c), (d), (e), (f) and (g), automatically forwarding, by the web portal, to the intelligent subsystem of the first user, in near real time, one or more sale offers to purchase the product or the service, from the selected seller; (i) in response to at least (a), (b), (c), (d), (e), (f), (g) and (h), automatically accepting, by the web portal, like votes and dislike votes for the selected seller from the first user, the second user and a plurality of third users; (j) in response to at least (a), (b), (c), (d), (e), (f), (g), (h) and (i), automatically determining a number of the like votes and a number of the dislike votes; (k) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i) and (j), automatically determining a seller score for the selected seller based on the number of the like votes and the number of the dislike votes by an algebraic equation, a statistical method, a statistical method coupled with conjoint analysis, or an algorithm based on quorum sensing; (l) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i), (j) and (k), automatically displaying the seller score for the selected seller; and (m) in response to at least (a), (b), (c), (d), (e), (f), (g), (h), (i), (j), (k) and (l), performing at least one of (i) negotiating, by the selected seller with the first user, a price for the product or the service or (ii) picking, by the first user, the price for the product or the service, wherein the said method steps in (a), (b), (c), (d), (e), (f), (g), (h), (i), (j), (k), (l) and (m) are at least an ordered combination or in an ordered sequence.

16. The method according to claim 15, wherein at least one of the one or more bots is a multimodal bot, wherein the multimodal bot comprises one or more computer implementable interfaces to comprehend and react to a voice, a text and a visual input.

17. The method according to claim 15, wherein the web portal is coupled with one or more software agents.

18. The method according to claim 15, wherein the product, the service, or a service contract is coupled with a blockchain.

19. The method according to claim 15, wherein the web portal is receiving an input data from the second user, a near-field communication (NFC) tag, a quick response (QR) code, or an object, wherein the object comprises a sensor and/or a wireless transmitter.

20. The method according to claim 15, further comprising the first user paying for the product or the service by transferring a currency or a bitcoin from the first user to the selected seller, to the second user, or to the plurality of third users by the web portal.